Standard of Camera & Imaging Products Association

CIPA DC-X011–Translation-2024

Measurement and Description Method for Image Stabilization Performance of Digital Cameras (Optical Systems)

This translation is based on the original Standard (CIPA DC-X011-2024). In the event of any doubts arising as to the contents, the original Standard in Japanese shall prevail.

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Revision History

Sept. 2012, Standard Established	CIPA Standard DC-011-Translation-2012	- Measurement and Description Method for Image Stabilization Performance of Digital Cameras defined.
Feb. 2015, Standard Revised	CIPA Standard DC-011-Translation-2015	- Added rules for notation on image stabilization performance for cameras with image stabilization for directions other than yaw and pitch. - Supplemental explanations added and format adjusted throughout the standard.
Jan. 2024, Standard Revised	CIPA Standard DC-X011–2024	- Changed handheld blur determination level to 20 μm. - Added roll handheld blur waveform to vibration waveforms. - Added image stabilization performance measurement at image height 60%. - Added supplemental explanations throughout the standard.

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1. Introduction

As digital cameras have permeated the market in recent years, new functions have successively been incorporated into new cameras to make them more convenient for users. One such function commercialized in many cameras is image stabilization, which detects camera shake and corrects image blur. The advantages of image stabilization performance are presented in brochures for equipped products. To date, however, the industry has not developed any unified methods for measuring or describing the image stabilization performance of cameras. This has made it difficult to compare image stabilization performance between cameras based on their brochures.

To date, the Camera and Imaging Products Association (CIPA) has established standards and guidelines for digital cameras, battery life, resolution, sensitivity and brochure descriptions. Brochures of many digital cameras are prepared in compliance with these standards and guidelines, making for fairer competition.

In the context of this background, this standard in 2012 added measurement and description methods for image stabilization performance of digital cameras to existing standards and guidelines. In the 2015 revision, the standard added description methods for image stabilization performance for cameras with image stabilization in directions other than yaw and pitch. In 2024, in response to subsequent changes in photographic viewing environments, this standard is revised to allow for measurement and description of roll image stabilization performance in addition to yaw and pitch.

This standard specifies the methods with regard to image stabilization as a type of blur suppression as defined in 3-1-2; however, it does not specify methods with regard to blur reduction or motion blur correction.

Typical systems of image stabilization include optical and electronic systems. In both systems, exposure time (nearly equal to shutter speed) is an important factor. Electronic image stabilization, however, has diverged into a broad spectrum of systems, many of which do not have clear definition of exposure time, making it difficult to measure image stabilization performance on the basis of this standard.

Hence, electronic image stabilization systems and hybrid image stabilization systems that use both optical and electronic image stabilization are excluded from the application of this standard as measurements are highly likely to be questionable.

In addition, for the purpose of supplementing the contents of this standard, an explanation of terminology relating to "handheld blur" is provided in the appendix.

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2. Scope

This standard applies to digital cameras for consumers with optical image stabilization for still image shooting, with the following provisions:

- Professional or industrial applications in which users and suppliers arrange case-by-case specifications are excluded.
- Applications with electronic image stabilization systems and with hybrid image stabilization systems that use both electronic and optical image stabilization are excluded.
- Apparatuses such as camcorders and mobile phones with still image shooting ability shall not be precluded from adopting this standard.

This standard is mainly meant for application in describing image stabilization performance in brochures or other printed media and websites that specify the product. However, the objectives of this standard are also to be complied with in notation on camera bodies and individual packages, as well as expressions used in advertising campaigns and point-of-purchase advertising for sales promotions.

Furthermore, any party other than suppliers taking measurements based on this standard (hereinafter called “measurer”) shall not be precluded from publishing the results of their measurements in their own publication.

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3. Definition of Terms and Referenced Standards

3-1 Definition of Terms

The terms used in this standard are defined as given below.

3-1-1 Blur

"Blur" means deviation from a predetermined position. Used as a camera term, blur is broadly divided into handheld blur and motion blur.

1) Handheld Blur

“Handheld blur” refers to a movement of a camera caused by unstable hands holding the camera, resulting in blurring of a subject in an image taken.

2) Motion Blur

“Motion blur” refers to blurring of a subject in an image caused by movement of the subject or a part of the subject during exposure, irrespective of any handheld blur.

3-1-2 Blur Suppression

“Blur suppression” refers to suppression of handheld blur or motion blur by some means of control within a camera. Although blur suppression can be further classified into image stabilization and blur reduction, this standard excludes applications for blur reduction.

Additionally, while image stabilization can be further classified into handheld blur correction and motion blur correction, this standard excludes applications for motion blur correction.

1) Image Stabilization

This function corrects for handheld blur in the output image as caused by movement of the camera body due to handheld blur (see 3-3-6) by using output by the blur detection device.

However, methods ultimately meant primarily to increase shutter speed by optimizing an exposure control program are considered "blur reduction" as described below, even when using output by the blur detection device.

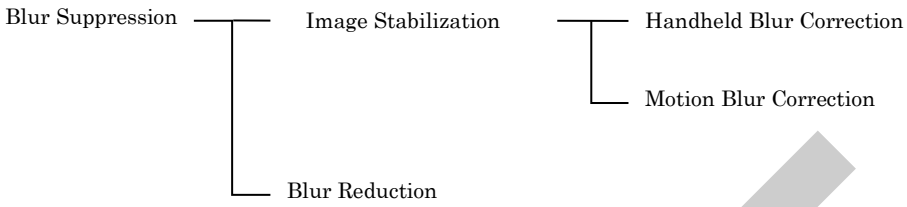
2) Blur Reduction

“Blur reduction” is a function or mode for reducing handheld blur in an output image as caused by either subject movement or handheld blur primarily by optimizing an exposure control program to increase shutter speed.

Image stabilization and blur reduction shall not be confused, as there are great differences in their

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comprising technical elements.



3-1-3 Vibratory apparatus

A “vibratory apparatus” is an apparatus capable of generating vibration in response to the amplitude of input vibration waveforms. Typically, the camera to be measured is mounted on the movable part of the vibratory apparatus using given fixing methods to transmit the vibration to it.

3-1-4 Vibration Waveform

A “vibration waveform” is an embodiment of camera shake (vibration) that occurs when shooting a subject with a camera by temporally shifting the amplitude value. A vibration waveform adopted in this standard has three axes: yaw and pitch, which are vibration components especially dominant in an image taken, and roll, which is a vibration component having the next most significant impact.

As the characteristics of camera shake vary depending on camera mass, two types of vibration waveforms shall be employed: WB-H and WB-L. (For the definition of "weight," see CIPA DCG-005 Measurement and Description Methods of Weight and Dimensions of Digital Cameras.)

The WB-H and WB-L files are in text format, named WB-H-xxx.txt and WB-L-xxx.txt, respectively. The waveforms are provided as angle data changing in the time direction for a length of 32 seconds. Data sampling frequency is 500Hz. The ‘xxx’ in the waveform file names is a 3-digit number giving the waveform version. If a waveform has been revised, version differences are managed with these numbers.

3-1-5 Handheld Blur Amount

“Handheld blur amount” is an amount that corresponds to shift of a subject in an image taken as induced by handheld blur.

3-1-6 Image Degradation and Image Degradation Amount

“Image degradation” refers to a phenomenon where deviation between the focal plane of a lens and image plane of the image sensor or where handheld blur deteriorates the sharpness in an image taken.

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Image degradation is sometimes caused by image processing of image data. “Image degradation amount” is a quantified value of the magnitude of image degradation.

3-1-7 Handheld Blur Measurement Chart

A “handheld blur measurement chart” is a chart to be used as a subject in measuring image stabilization performance. When establishing this standard, CIPA prepared a handheld blur measurement chart called the “CIPA handheld blur measurement chart.”

3-1-8 Handheld Blur Measurement Software

"Handheld blur measurement software" is provided by CIPA and dedicated software for measuring the intrinsic image degradation amount (3-1-10) and the total image degradation amount (3-1-14) from a captured image using the CIPA handheld blur measurement chart (3-1-7). “Pixel” is the unit for the image degradation amount generated by this software.

3-1-9 Stop (Stop Number)

According to APEX expressions, the difference in TV values is represented in ”stops.” For instance, the difference between shutter speeds of 1/1000 (TV10) and 1/500 seconds (TV9) or that between shutter speeds of 1/125 (TV7) and 1/60 seconds (TV6) accounts for one stop.

3-1-10 Intrinsic image degradation amount

“Intrinsic image degradation amount” is the image degradation amount of an image taken caused by factors other than handheld blur. Intrinsic image degradation amount is unique to an individual instrument and is dependent on the optical performance, effective number of pixels and image processing of the camera to be measured. By removing intrinsic image degradation amount, almost all factors affecting image stabilization performance other than the image stabilization function can be removed.

3-1-11 Determination Level for Image Stabilization Performance

“Determination level for image stabilization performance” is the level used in judgments of image stabilization performance calculations. In this standard, this level is defined as 20 μm.

3-1-12 Theoretical Handheld Blur Amount

“Theoretical handheld blur amount” is a theoretical value for the amount of measurable handheld blur in an image taken with image stabilization turned OFF (the function is assumed to be OFF for cameras without an OFF setting) when the camera to be measured is vibrated with a vibration waveform.

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3-1-13 Reference Image Degradation Amount

“Reference image degradation amount” is the theoretical estimated image degradation amount for an image taken with image stabilization turned OFF (the function is assumed to be OFF for cameras without an OFF setting) when the camera to be measured is vibrated with a vibration waveform. This amount is expressed as the square root of the sum of the squares of the intrinsic image degradation amount and the theoretical handheld blur amount.

3-1-14 Total Image Degradation Amount

“Total image degradation amount” is the actual measurement value of image degradation amount for an image taken with image stabilization turned ON when the camera to be measured is vibrated with a vibration waveform.

3-1-15 Reference Handheld Blur Amount

“Reference handheld blur amount” is a reference value for use in calculating image stabilization performance. The reference handheld blur amount is calculated by subtracting the intrinsic image degradation amount from the reference image degradation amount caused when the camera is vibrated for cameras not equipped with image stabilization or with image stabilization turned OFF (the function is assumed to be OFF for cameras without an OFF setting).

3-1-16 Measured Handheld Blur Amount

“Measured handheld blur amount” is a value representing residual handheld blur after correction with image stabilization of the camera to be measured ON. This value is calculated by subtracting intrinsic image degradation amount from total image degradation amount.

3-1-17 Average Vibration Angle

“Average vibration angle” is the average value for vibration angles caused by vibration with a vibration waveform (3-1-4) at each shutter speed.

Average vibration angle data files are in text format, with the file corresponding to WB-H named AVA-1-xxx.txt and that corresponding to WB-L named AVA-2-xxx.txt. The file corresponding to Selection Criteria III described in 4-2-6 is named AVA-3-xxx.txt.

Again, the ‘xxx’ in the file names is a 3-digit number giving the version number. The 3 digits are used to manage the files, corresponding with the waveform revisions.

3-1-18 Shutter Speed

“Shutter speed” is a theoretical exposure time value.

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In this standard as well, shutter speeds are denoted using typical display values (30, 15, 8, 4, 2, 1, 1/2, 1/4, 1/8, 1/15, 1/30, 1/60, 1/125, 1/250, 1/500, 1/1000, etc.). At the time of calculation, real numbers (32, 16, 8, 4, 2, 1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, 1/128, 1/256, 1/512, 1/1024, etc.) or values based on real numbers shall be used. (Refer to CIPA DCG-008-2022 Guideline for Presentation Method of Shutter Speed on Digital Cameras.)

3-1-19 Image Clipping Software

"Image Clipping Software" is dedicated software provided by CIPA that partially clips a captured image using the CIPA handheld blur measurement chart (3-1-7) at the center of the image and four positions at image height 60% in diagonal directions, storing each partial image in five different folders. With the images stored in these five folders, intrinsic image degradation amount (3-1-10) and total image degradation amount (3-1-14) are measured by the handheld blur measurement software (3-1-8), as necessary.

3-2 Referenced Standards

3-2-1 Referenced Standards

- CIPA DCG-002 Specification Guideline for digital cameras
- CIPA DCG-005 Measurement and description methods of weight and dimensions of digital cameras
- CIPA DCG-008-2022 Guideline for Presentation Method of Shutter Speed on Digital Cameras

3-2-2 Response to Revision of Reference Standard

If the standards and guidelines referred to in this standard are revised, compliance shall be with the revised editions.

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4. Measurement Method

4-1 Preconditions

The objective of this standard is to specify how to measure image stabilization performance in the case of users taking a shot with the camera in hand. Accordingly, measuring sessions will better simulate real shooting if the camera to be measured is actually held by hand. This does, however, make it difficult to eliminate variations due to differences among individual photographers or how well the camera is designed for handheld shooting. In order to ensure fair measurement results, the camera to be measured is to be mounted on a vibratory apparatus to vibrate the camera with a vibration waveform to be discussed later, and image stabilization performance shall be measured on the basis of results of shooting the handheld blur measurement chart specified in this standard.

Information on how to get information and tools (vibration waveforms, handheld blur measurement software, CIPA handheld blur measurement chart, etc.) additionally required for the measurement of image stabilization performance as specified in this standard shall be separately presented by CIPA.

Measurements of image stabilization performance shall comply with the following: 4-2 Equipment and Environment for Measurement, 4-3 Setting of Camera to be Measured and 4-4 Measurement Procedures. Determinations on the number of cameras to be measured is left to the supplier or the measurer, as long as the number does not render questionable measurement results. Figure 4-1 shows the overview of the measurement method.

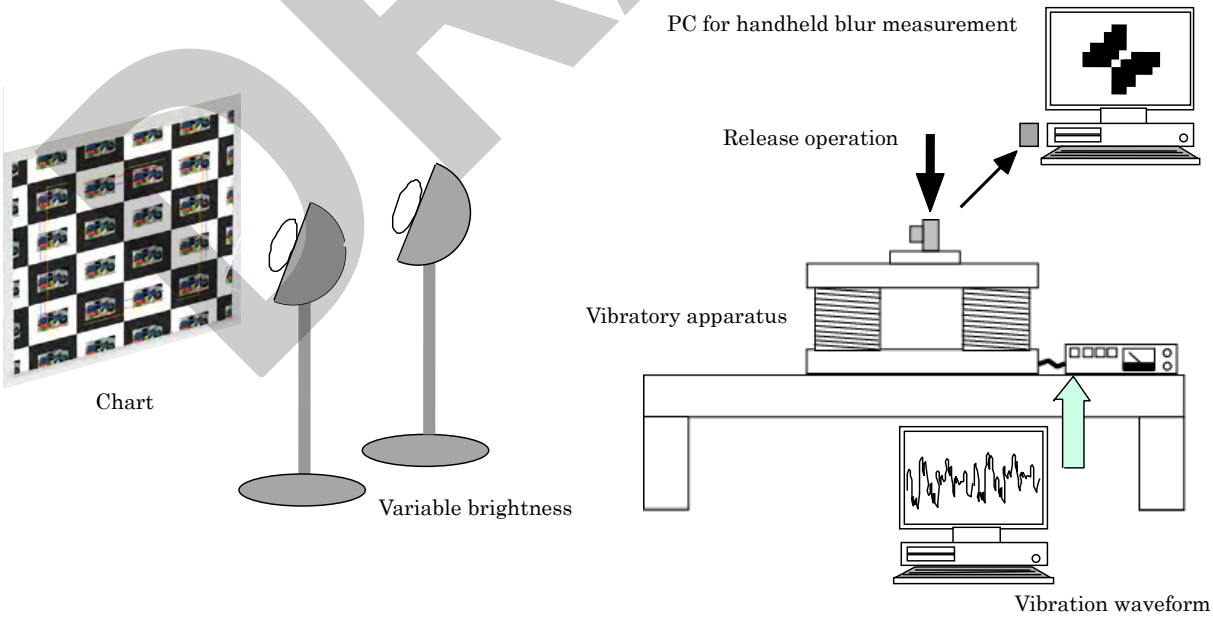


Figure 4-1 Overview of Measurement Method

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4-2 Equipment and Environment for Measurement

4-2-1 Handheld Blur Measurement Chart

For this standard, a handheld blur measurement chart satisfying the following specifications shall be used for the measurement of image stabilization performance.

1) The chart shall be comprised of measurement sites at the center of the image and positions at image height 60%, and may optionally include a portion having natural color images.
2) The ratio of reflectance between the white and black regions of the measurement sites shall be 4:1 or higher, and the white and black region shall be wider than the expected maximum image degradation amount.
3) No particular features are specified for the natural image portion, but the portion preferably contains images similar to real subjects.

Use of the CIPA handheld blur measurement chart illustrated in Figure 4-2-1a is recommended. The CIPA handheld blur measurement chart has natural color images inserted to facilitate focusing or general purpose applications. The outline dimensions of the CIPA handheld blur measurement chart are 1000 mm × 750 mm in consideration of shooting magnification.

White and black edge areas for measurement are provided at the center and at four locations at image height 60% within the frames indicating the shooting range. To ensure measurement at corner positions of image height 60% even with differing aspect ratios of imaging elements, the chart has three frames (aspect ratios of 3:2, 4:3, and 16:9); therefore, the camera's shooting range shall be adjusted in accordance with the format of the imaging element so that the frame lines substantially match the screen edges to be shot. The lens focal length shall then be determined and the shooting distance adjusted. This process is assumed in the descriptions of this standard below.

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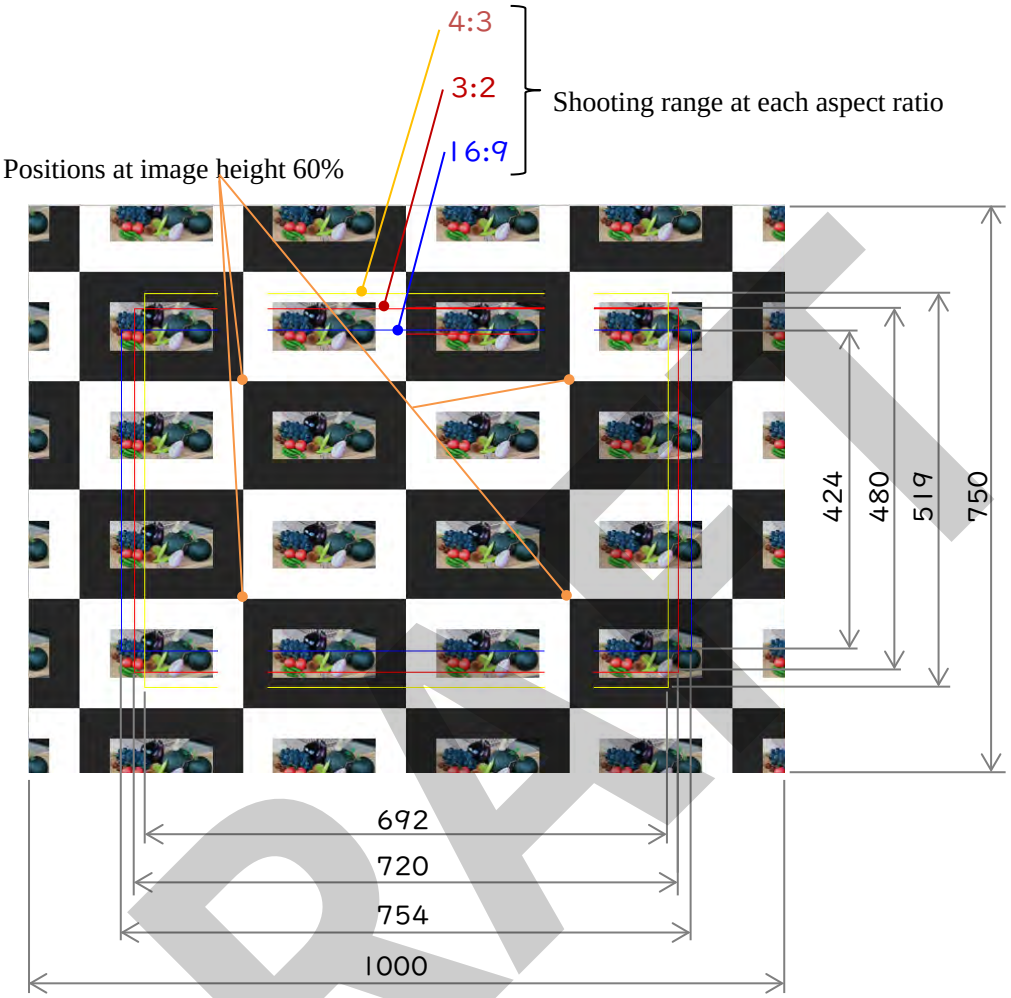


Figure 4-2-1a CIPA handheld blur measurement chart (unit: mm)

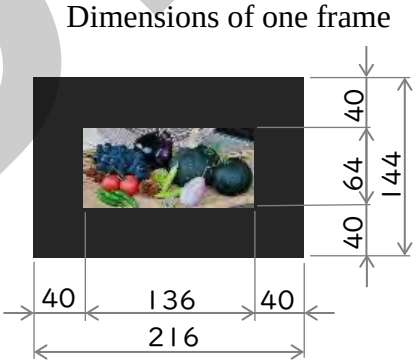


Figure 4-2-1b: One frame of chart (unit: mm)

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4-2-2 Lighting

Although no specific types of lighting are defined, lighting shall not cause flicker in the image during shooting. Measures should also be taken to prevent reflection from the light source or uneven illuminance on the handheld blur measurement chart to prevent any question relating to measurement results.

Lighting with variable brightness is useful.

4-2-3 Temperature and Humidity

The temperature and humidity shall be 23°C ± 2°C and 30 to 70%. Measurements may be performed under other conditions as long as there is no room for question.

4-2-4 Vibratory Apparatus

The vibratory apparatus to use shall function in a manner that accurately reproduces vibrations according to the vibration waveforms. Always verify that the vibratory apparatus is vibrating accurately before taking measurements. To do so, measurers of image stabilization performance or manufacturers of vibratory apparatuses refer to **Verification Method of Vibratory Apparatus** specified separately.

4-2-5 Mounting of Camera to be Measured on Vibratory Apparatus

When mounting the camera to be measured on the vibratory apparatus, vibration of the vibratory apparatus and that of the camera to be measured mounted on the vibratory apparatus have to match.

When measuring a camera with a long-barrel lens (e.g. high-powered zoom lens), vibrations of the camera body and the lens may not match due to distortion induced in the lens by excitation preventing the applied vibration from being correctly transmitted to the lens. Thus, given measures are to be taken for such measurements so that lens and camera body vibrations match, such as fixing the lens to the vibratory apparatus in addition to the camera body.

The vibration waveforms adopted in this standard are vibrations that rotate on the rotational axes of yaw, pitch, and roll; therefore, the positional relationship between the center of rotation of the vibratory apparatus and the optical axis of the camera and lens to be measured at the time of mounting affects the vibration measurement results. (Hereafter, camera and lens are abbreviated as camera.)

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Therefore, to ensure measurement condition alignment, the center of rotation of the vibration table are preferably aligned with the optical axis of the lens in the horizontal and vertical directions on the plane orthogonal to the optical axis.

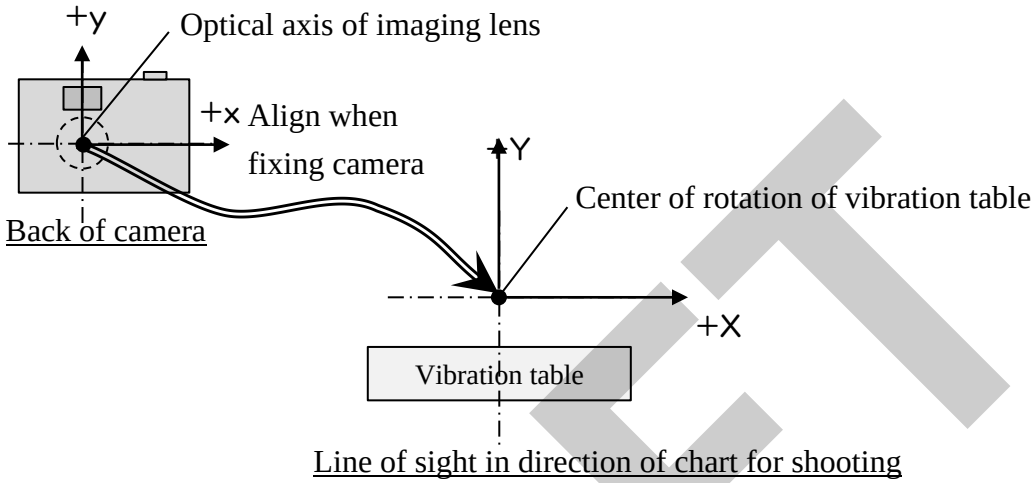


Figure 4-2-5a Center of rotation of vibration table and optical axis of lens

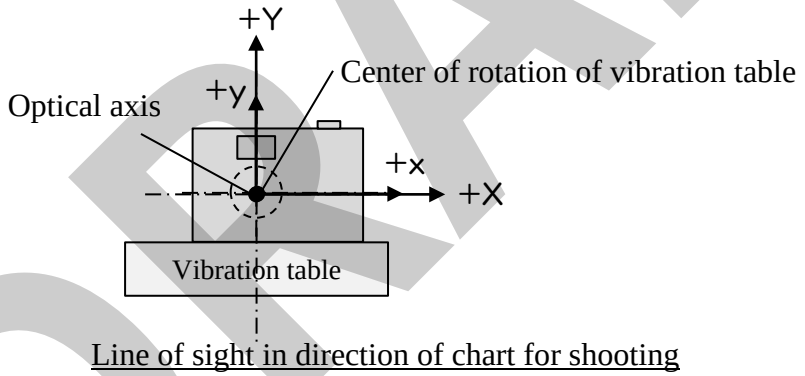


Figure 4-2-5b Camera fixed with lens optical axis aligned with center of rotation of vibration table

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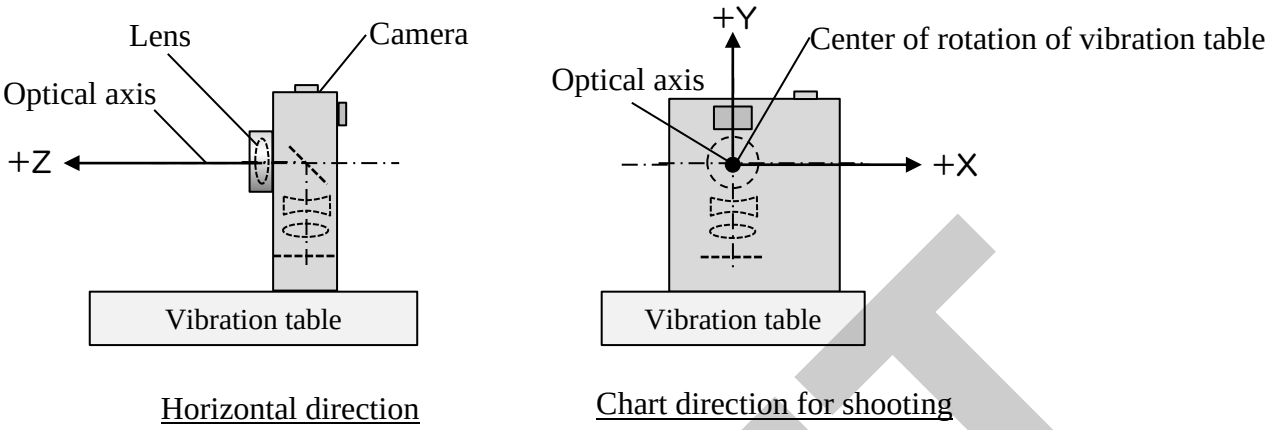


Figure 4-2-5c Fixed state of camera with bending optical system

4-2-6 Vibration Waveform

There are two types of vibration waveforms: WB-L and WB-H. Criteria for selecting a waveform are as follows: (Here, total mass is the sum of the masses of a camera body, the included storage media and battery, and a lens)

Selection Criterion I. WB-H shall be used for total mass of 600 g or more;

Selection Criterion II. WB-L shall be used for total mass of less than 400 g;

Selection Criterion III. Both WB-L and WB-H shall be used for total mass between 400 g and 600 g.

Both waveforms consist of 3-axis components: yaw, pitch, and roll. Yaw, pitch, and roll components shall be excited at the same time.

(Note: Given that camera geometries and shooting methods change and image stabilization techniques will be improved in response to such changes, the vibration waveforms specified in this standard are subject to revision as necessary.)

4-2-7 Shooting Distance

When the vibratory apparatus is stopped, the shooting distance shall be adjusted so that the white and black edges at image height 60% at the four positions in the CIPA handheld blur measurement chart in Figure 4-2-1a are within an image height of 60 ± 3%. This distance is 20 times the 35mm film equivalent focal length. The objective of the adjustment is to ensure that the image is taken so that the range of the frame indicated for each aspect ratio depicted in Figure 4-2-1a CIPA Handheld Blur Measurement Chart covers the entire screen area when the vibratory apparatus is stopped.

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When using a handheld blur measurement chart other than the CIPA handheld blur measurement chart, the measurement shall be made after confirmation that the measurement positions at image height 60% are within image height 60 ± 3%.

For example, when measurement is made with a camera having an aspect ratio of 3:2, the image shall be taken with the range of the 3:2 brown frame in Figure 4-2-1a CIPA Handheld blur Measurement Chart filling the entire screen. When taken with the imaging range properly filling the screen, the image is taken within the red virtual solid line in Figure 4-2-7 Shift in Shooting Range and Change in Image Height at Corners. In this case, the white and black edges at image height 60% at the four locations on the chart are at image height 60% on the image.

With the shooting distance farther away and the image taken only within the purple virtual solid line in Figure 4-2-7, the white and black edges at image height 60% at the four locations on the chart are at image height 57% on the image. Conversely, with the shooting distance closer and the image taken only within the green virtual solid line in Figure 4-2-7, the white and black edges at image height 60% at the four locations on the chart are at image height 63% on the image.

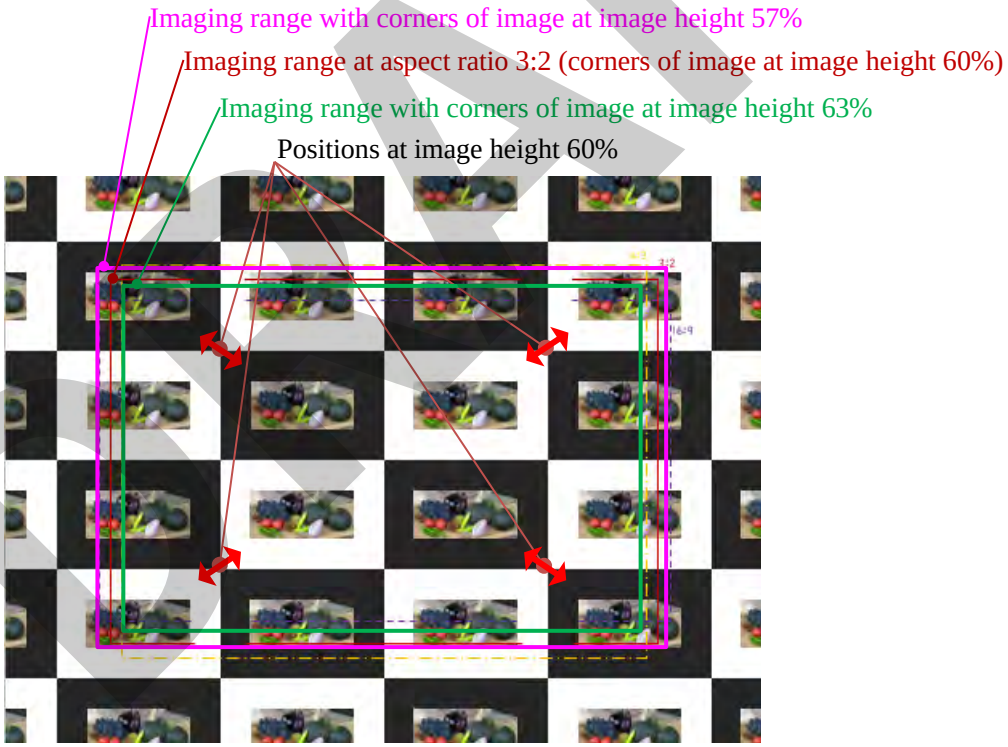


Figure 4-2-7 Shift in Shooting Range and Change in Image Height at Corners

4-2-8 Handheld Blur Measurement Software

Handheld blur measurement software (3-1-8) can be used to measure the intrinsic image degradation amount and total image degradation amount of an image of the CIPA handheld blur measurement chart

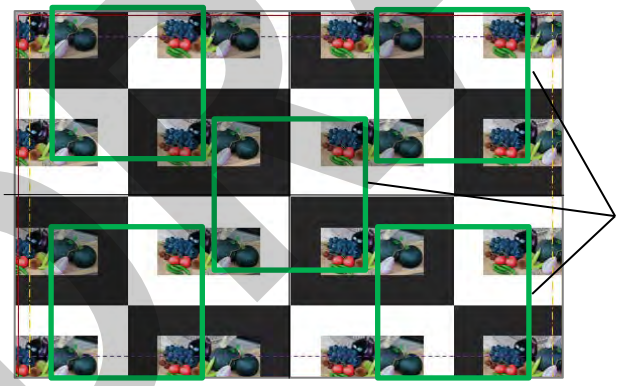
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as taken in accordance with 4-4 Measurement Procedures. This software generates an intrinsic image degradation amount and a total image degradation amount in pixels. Use of this software is recommended for measuring intrinsic image degradation amount and total image degradation amounts for this standard. The CIPA handheld blur measurement chart shall be used without fail when this software is used.

Unreasonable measurement results may be generated even when using this software depending on the settings of the camera to be measured or measurement environment. If measurement results are even slightly questionable, review the settings of the camera to be measured or the measurement environment and take measures such as repeating the measurement.

This handheld blur measurement software is specified to calculate the image degradation amount at the white and black edges near the center of the image. Therefore, as a pre-processing step, the captured image, centering the clips about the areas subject to image degradation amount calculation, shall be clipped. Specifically, with use of the image clipping software separately provided by CIPA, portions of the image of the CIPA handheld blur measurement chart corresponding to the center and four positions in diagonal directions at image height 60% are clipped, and these partial images are stored in five different folders. This image clipping process may be performed using in-house software.

Each image degradation amount is then calculated from the images in each folder with the handheld blur measurement software.



4-2-8a Example of Image Clipping Range of Five Locations from Captured Image

If a method other than the handheld blur measurement software specified in this standard is used, the intrinsic image degradation amount and total image degradation amount shall be measured in accordance with the following:

1) The signal levels of the boundary regions between black portions and white portions in the image of a shot chart shall be normalized from 0% to 100%. To eliminate the impact of noise, the number of pixels in the section between 10% to 90% of signal levels (A in Figure 4-2-8) shall be calculated

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and then multiplied by 10/8;

2) At this time, the tone characteristic shall be linearized to the extent possible. (Before normalization, the signal level is tone corrected.)
3) For stable measurements, the boundary regions between black portions and white portions which are the measurement locations of the image of the shooting chart shall be used.

Refer to the handheld blur measurement software instruction manual for details on how to measure intrinsic image degradation amount and total image degradation amount using this software.

Any variation or error in measurement of the intrinsic image degradation amount and total image degradation amount (or measured handheld blur amount) would compromise the image stabilization performance measurements for this standard. As such, the accuracy of the measured intrinsic image degradation amount, comprehensive image degradation amount and measured handheld blur amount shall be verified without fail before starting the measurement.

Refer to the separately specified Verification Method of Vibratory Apparatus for methods of verifying measurement.

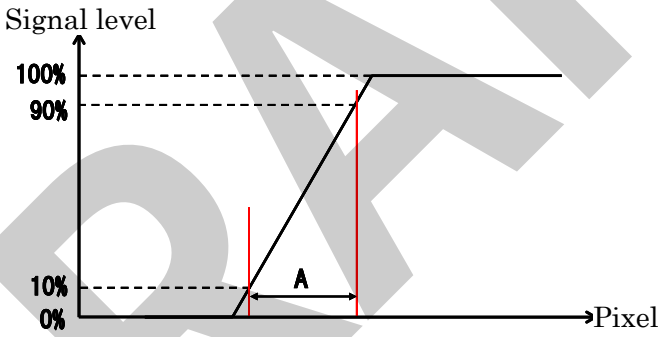


Figure 4-2-8b Measuring intrinsic image degradation amount and total image degradation amount

4-3 Settings of Camera to be Measured

4-3-1 Power to be Used

Not specified. Cameras capable of using external power sources may use external power.

4-3-2 Shooting Mode

No specific shooting mode is stipulated. However, given that shooting shall begin within three seconds after shooting is enabled following power ON, preferable shooting modes will either be capable of remembering as many of the items specified in 4-3-3 to 4-3-20 as possible and setting them

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in advance with the power OFF or allow for easy changing of the shutter speed. Note that the shooting mode should not include any image processing which might produce questionable measurement results.

As an example, many cameras have modes adapted to specific shooting scenes. Many such modes include image processing that is optimized for given situations. This introduces the possibility of questionable measurement results, making such modes unsuitable for image stabilization performance measurement. Accordingly, such modes adapted to specific situations which include features such as extreme edge enhancement shall not be used in principle, even if the mode includes allowing for easy changing of the shutter speed.

4-3-3 Image Stabilization Mode

Setting shall be in the image stabilization mode to be measured.

4-3-4 Image Quality Mode (Compression Ratio)

Although no specific compression ratio is specified, a high image quality mode setting with low compression ratio is preferable.

4-3-5 Image Quality Mode (Number of Recording Pixels)

Although no number is specified for recorded pixels, this will preferably be set to the maximum number of recording pixels for the camera to be measured. However, settings that would exceed the effective number of pixels of the image sensor by means of pixel interpolation, image processing or other alterations shall not be adopted.

4-3-6 ISO Sensitivity

ISO sensitivity shall be set to minimize any image noise. If possible, ISO sensitivity should be set to a constant value.

4-3-7 Flash

Not to be used.

4-3-8 Storage Media Capacity

Not specified.

4-3-9 Focal Length for Measurement (Cameras Mounted with Zoom Lenses)

Not specified.

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4-3-10 Electronic (Digital) Zoom

Not used.

4-3-11 Focus Control

No focus control system, such as autofocus and manual focus, is specified. However, the camera shall be correctly focused on the handheld blur measurement chart.

4-3-12 White Balance

White balance shall be appropriate to the light sources used.

4-3-13 Color Setting

Although no specific color setting is specified, the factory default setting is preferable.

4-3-14 Exposure Setting

Although no specific exposure setting is specified, the camera shall be set such that it can shoot at an appropriate exposure.

4-3-15 Aperture

Although no specific aperture value is specified, aperture shall be kept constant if possible when shooting at the same focal length and the same shutter speed.

4-3-16 Aspect Ratio

Although no specific aspect ratio is specified, the factory default setting is preferable.

4-3-17 Continuous Shooting Mode

Although no specific continuous shooting mode is specified, camera settings should allow performance of 4-4-1 Measurement of Intrinsic image degradation amount and 4-4-2 and 4-4-3 Measurement of Total Image Degradation Amount as discussed below.

4-3-18 Self Timer

Although no specific self-timer setting is specified, camera settings should allow performance of 4-4-1 Measurement of Intrinsic Image Degradation Amount and 4-4-2 and 4-4-3 Measurement of Total Image Degradation Amount as discussed below.

4-3-19 Aberration Correction Setting

No specific aberration correction settings are specified, but factory default settings are preferable.

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4-3-20 Other

When shooting at differing shutter speeds, preferably no measurement conditions other than lighting will be changed. Conditions that may result in questionable measurement results shall be avoided.

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4-4 Measurement Procedures

4-4-1 Measurement of Intrinsic image degradation amount:

1) Make preparations on the basis of environment settings described in 4-2 and 4-3;
2) Mount the camera to be measured on the vibratory apparatus;
3) Turn the power of the camera to be measured ON. For cameras mounted with a zoom lens, set the lens to the focal length for image stabilization performance measurement. Here, in accordance with the image aspect ratio of the camera to be measured, adjust the shooting distance and the position of the camera to be measured or the position of the CIPA handheld blur measurement chart so that the portion enclosed by the frame indicating the corresponding aspect ratio in Figure 4-2-1a fills the entire screen during shooting;
4) The shutter speed shall be set around ”1/ focal length (35 mm film equivalent)”;
5) Start shooting with the vibratory apparatus stopped, shooting at least ten images. While there is no specified upper limit for number of images taken, all images shall be used without selecting only those that are desirable. Image stabilization is to be turned OFF, unless there is no OFF setting. It is recommended to use a remote release, remote control or similar device if possible.
6) Reduce the shutter speed sequentially by at most one stop at a time. Continue shooting in like fashion until the necessary and sufficient shutter speed is attained.
7) The intrinsic image degradation amount shall be measured at the center and, if a performance description at each corner of the image is required, at each of the four positions at image height 60%.

The shooting environment and camera settings in 4-4-1 are generally not to be changed from the Measurement of the Total Image Degradation Amount of 4-4-2 and 4-4-3.

4-4-2 Measurement of Total Image Degradation Amount (for 4-2-6 Selection Criteria I and II):

1) Make preparations on the basis of the environment setup shown in 4-2 and 4-3;
2) Mount the camera to be measured on the vibratory apparatus;
3) Turn the power of the camera to be measured ON. For cameras mounted with a zoom lens, set the lens to the focal length for image stabilization performance measurement. Here, in accordance with the image aspect ratio of the camera to be measured, confirm, if necessary, the shooting distance as well as the position of the camera to be measured or the position of the CIPA handheld blur measurement chart so that the portion enclosed by the frame indicating the corresponding aspect ratio in Figure 4-2-1a fills the entire screen during shooting; Further, shutter speed shall be set such that the measured handheld blur amount is estimated to be at or around the determination level for image stabilization performance mentioned below. After setting is completed, turn the power OFF once;

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4) Activate the vibratory apparatus using the vibration waveform stipulated in 4-2-6. Here, the vibratory apparatus should remain continuously activated until Step 8) below is completed. If the vibratory apparatus deactivates in the middle due to its specification, resume vibration immediately;

5) While vibrating, turn the power of the camera to be measured ON again at unspecified timing. If the settings in 3) above are changed, reset them as soon as possible;

6) After steps 1) to 5) above are completed, with image stabilization ON for the camera to be measured, turn the power ON and, once shooting is enabled, start shooting within three seconds. Images shall be shot at an interval of approximately one second for a total of 10 shots. Turn the power OFF after shooting is finished;

7) Repeat the Steps 5) and 6) and take 200 or more shots. While there is no specified upper limit for number of images taken, all images shall be used without selecting only those that are desirable. Note that the shooting operation is not to affect the vibrating state. Where permitted, use of a remote release or a remote control is desirable;

8) Reduce or increase the shutter speed by at most one stop at a time and repeat Steps 3) to 7) above; and

9) Finish measurement when two measurement results are obtained: the slowest shutter speed at which the measured handheld blur amount does not exceed the determination level for image stabilization performance, and the fastest shutter speed at which it exceeds the determination level for image stabilization performance.

4-4-3 Measurement of Total Image Degradation Amount (for 4-2-6 Selection Criterion III):

1) Make preparations on the basis of the shooting environment given in 4-2 and 4-3;

2) Mount the camera to be measured on the vibratory apparatus;

3) Turn the power of the camera to be measured ON. For cameras mounted with a zoom lens, set the lens to the focal length for image stabilization performance measurement. Here, in accordance with the image aspect ratio of the camera to be measured, confirm, if necessary, the shooting distance as well as the position of the camera to be measured or the position of the CIPA handheld blur measurement chart so that the portion enclosed by the frame indicating the corresponding aspect ratio in Figure 4-2-1a fills the entire screen during shooting; Further, shutter speed shall be set such that the measured handheld blur amount is estimated to be at or around the determination level for image stabilization performance mentioned below. After setting is completed, turn the power OFF once;

4) Activate the vibratory apparatus using the vibration waveform stipulated in 4-2-6. Here, the vibratory apparatus should remain continuously activated until Step 8) below is completed. If the vibratory apparatus deactivates in the middle due to its specification, resume vibration

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immediately;

5) While vibrating, turn the power of the camera to be measured ON again at unspecified timing. If the settings in 3) above are changed, reset them as soon as possible;
6) After steps 1) to 5) above are completed, with image stabilization ON for the camera to be measured, turn the power ON and, once shooting is enabled, start shooting within three seconds. Images shall be shot at an interval of approximately one second for a total of 10 shots. Turn the power OFF after shooting is finished;
7) Repeat Steps 5) and 6) using two kinds of waveforms, and take 100 or more shots for each waveform. While there is no specified upper limit for number of images taken, all images shall be used without selecting only those that are desirable. Note that the shooting operation is not to affect the vibrating state. Where permitted, use of a remote release or a remote control is desirable;
8) Reduce or increase the shutter speed by at most one stop at a time and repeat Steps 3) to 7) above; and
9) Finish measurement when two measurement results are obtained: the slowest shutter speed at which the measured handheld blur amount does not exceed the determination level for image stabilization performance, and the fastest shutter speed at which it exceeds the determination level for image stabilization performance.

4-4-4 Action when Total Image Degradation Amount at Image Height 60% Is Not Measurable

If, due to the long focal length of the lens, the position of image height 60% on the handheld blur measurement chart during excitation is outside the screen or close to the screen edge, making measurement not possible, exclude photographs not measurable at the position of image height 60% on the handheld blur measurement chart; then it is recommended to take additional photographs until at least 200 measurable photographs are obtained for selection criteria I and II, or additional photographs may be taken until at least 100 measurable photographs each are obtained for selection criteria III using two kinds of waveforms and to calculate the total image degradation amount using those photographs. However, if the number of additional shots is large, the measurement at image height 60% may be canceled. (Refer to the explanation provided in 6-26.)

4-5 Calculation of Image Stabilization Performance

4-5-1 Calculation of Basic Values Required to Calculate Image Stabilization Performance

Six quantitative values are used to calculate image stabilization performance: intrinsic image degradation amount, theoretical handheld blur amount, reference image degradation amount, measured image degradation amount, reference handheld blur amount, and measured handheld blur

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amount.

The main objectives of calculations are:

a) To remove the effects of image degradation, which is included in the total image degradation amount as measured in 4-4-2 and 4-4-3 and caused by factors other than handheld blur.

b) To allow for calculation of image stabilization performance according to this standard even for cameras incapable of turning image stabilization OFF.

For further detail, see ”6. Explanation.” Methods of calculating these values will be shown below.

1) Intrinsic image degradation amount

Intrinsic image degradation amount is quantified from the image taken according to the method described in 4-4-1. Specifically, the square root of the sum of the squares of the horizontal and vertical image degradation amounts as obtained from each image taken shall be calculated, and then the total of such values is averaged by the number of shots. If the averaged intrinsic image degradation amount is in pixels, it shall be converted into the 35 mm film equivalent intrinsic image degradation amount [μm] according to the method described in 4-5-2 below. However, values that are clearly questionable in the results of the image degradation amount measurement shall not be used. At least 10 values are to be used in calculating the average.

2) Theoretical handheld blur amount

Theoretical handheld blur amounts are to be calculated for each shutter speed using the following equation. Given the different average vibration angles for selection criteria I, II and III as described in 4-2-6 Vibration Waveform, note that the amount corresponding to the vibration waveform used in the measurement shall be employed. First, the yaw and pitch theoretical handheld blur amounts shall be calculated. This yaw/pitch theoretical handheld blur amount is proportional to the 35mm film equivalent focal length and the yaw/pitch average vibration angle.

Yaw/pitch theoretical handheld blur amount [μm] = 35mm film equivalent focal length f [mm] × tan θyp × 1000

θyp: Yaw/pitch average vibration angle

If the 35mm film equivalent focal length is unknown, the conversion factor (fr) may be used.

Yaw/pitch theoretical handheld blur amount [μm] = conversion factor (fr)　[mm] x tan

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θyp x1000

$$Conversion\ Factor\ (fr) = Shooting\ Distance\ (a)[m] \times \frac{Chart\ standard\ length\ (b)[pixels]}{Chard\ standard\ length\ (e)[mm]} \times \frac{\sqrt{24^2+36^2}[mm]}{\sqrt{Image\ width\ (c)^2+Image\ height\ (d)^2}[pixels]} \times 10^6$$

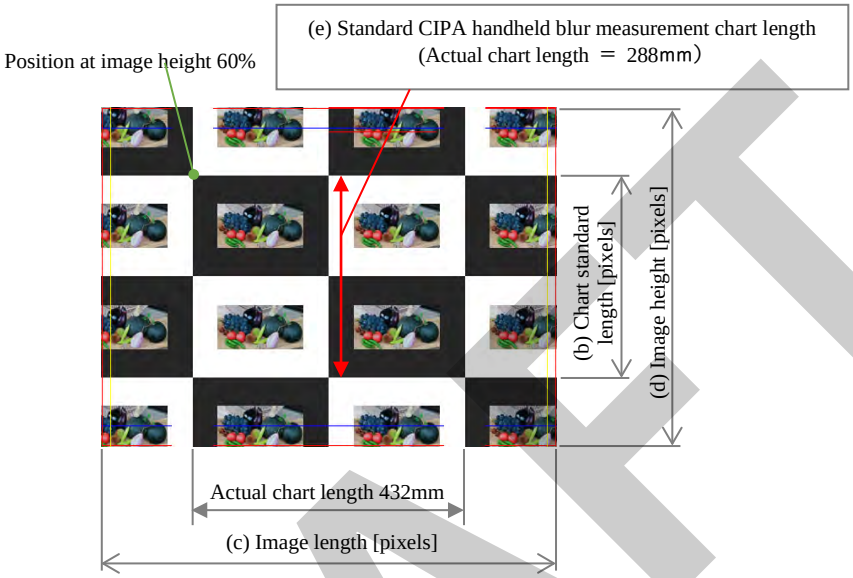


Figure 4-5-1a Each factor for obtaining the conversion factor (fr)

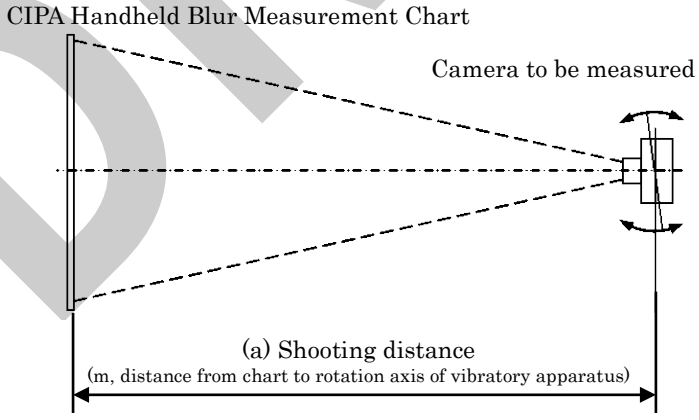


Figure 4-5-1b Shooting Distance (a)

Next, the roll theoretical handheld blur amount shall be calculated. This roll theoretical handheld

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blur amount is proportional to the 35mm film equivalent distance from the center of the image to the measurement point and the roll average handheld blur amount. The peripheral measurement points are positions at image height 60%. At the center, L = 0; therefore, the roll theoretical handheld blur amount is 0. Further, there is no relationship with the 35mm film equivalent focal length.

Roll theoretical handheld blur amount [μm] = 35mm film equivalent distance from center of image to measurement point L [mm] × tan θr × 1000

θr: Roll average vibration angle

L: 0mm at center, 12.980mm at 35mm film equivalent image height 60%.

Lastly, the theoretical handheld blur amount is obtained by the square root of the sum of the squares of the yaw/pitch theoretical handheld blur amount and the roll theoretical handheld blur amount.

Theoretical handheld blur amount [μm] =

$$\sqrt{\text{(Yaw/pitch theoretical handheld blur amounts [μm])}^2 + \text{(Roll theoretical handheld blur amount}}$$

3) Reference image degradation amount

Reference image degradation amount is calculated using the following equation. Note that values at the same shutter speed shall be used for the intrinsic image degradation amount and the theoretical handheld blur amount.

$$\text{Estimated Comprehensive Bokeh Amount[um]} = \sqrt{(\text{Bokeh Offset Amount[um]})^2 + (\text{Theoretical Motion Blur Amount[um]})^2}$$

4) Total Image Degradation Amount

Measured image degradation amount is quantified from the image taken according to the methods described in 4-4-2 and 4-4-3. Specifically, the square root of the sum of the squares of the horizontal and vertical image degradation amounts as obtained from each image taken shall be calculated, and then the total of such values is averaged by the number of shots. If the averaged intrinsic image degradation amount is in pixels, it shall be converted into the 35 mm film equivalent measured intrinsic image degradation amount [μm] according to the method described in 4-5-2 below.

This quantification shall be performed at the center and, if a performance description of image

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height at the corner of the image is required, performed at each of the four positions at image height 60% as well.

However, values that are clearly questionable in the results of the image degradation amount measurement shall not be used. At least 200 values shall be used to calculate averages for 4-4-2, and at least 100 shall be used for each case in 4-4-3.

5) Reference Handheld Blur Amount

Reference handheld blur amount is calculated using the following equation. Note that the reference image degradation amount and the intrinsic image degradation amount shall be values measured at the identical shutter speed. Factors that may affect images taken, such as ISO sensitivity, shall be identical.

Reference handheld blur amount [μm] = Reference image degradation amount [μm] − Intrinsic image degradation amount [μm]

6) Measured Handheld Blur Amount

Measured handheld blur amount is calculated using the following equation. Note that the total image degradation amount and the intrinsic image degradation amount are to be measured at the same shutter speed. Factors that may affect images taken, such as ISO sensitivity, shall be identical. If the measured handheld blur amount becomes a negative value, the value shall be 0 μm.

Measured Handheld Blur Amount [μm] = Total image degradation amount [μm] - intrinsic image degradation amount [μm]

This calculation shall be performed at the center and, if a performance description of image height at the corner of the image is required, performed at each of the four positions at image height 60% as well.

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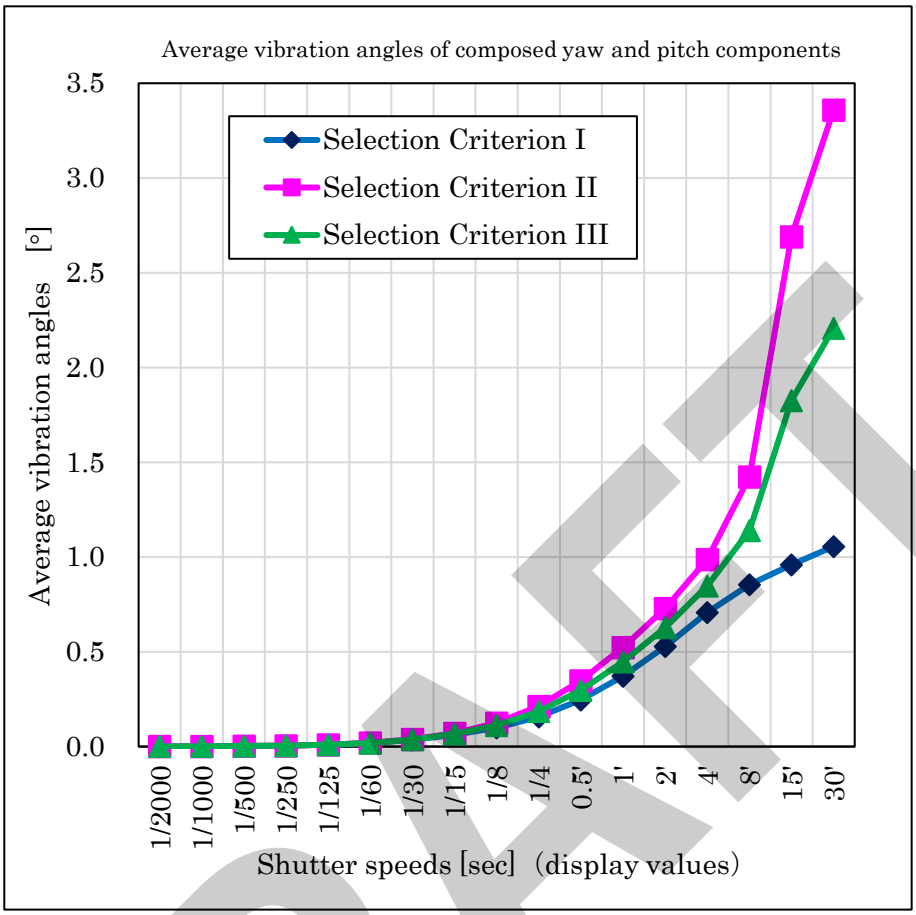


Figure 4-5-1c Average Vibration Angle of Composed Yaw/Pitch Component

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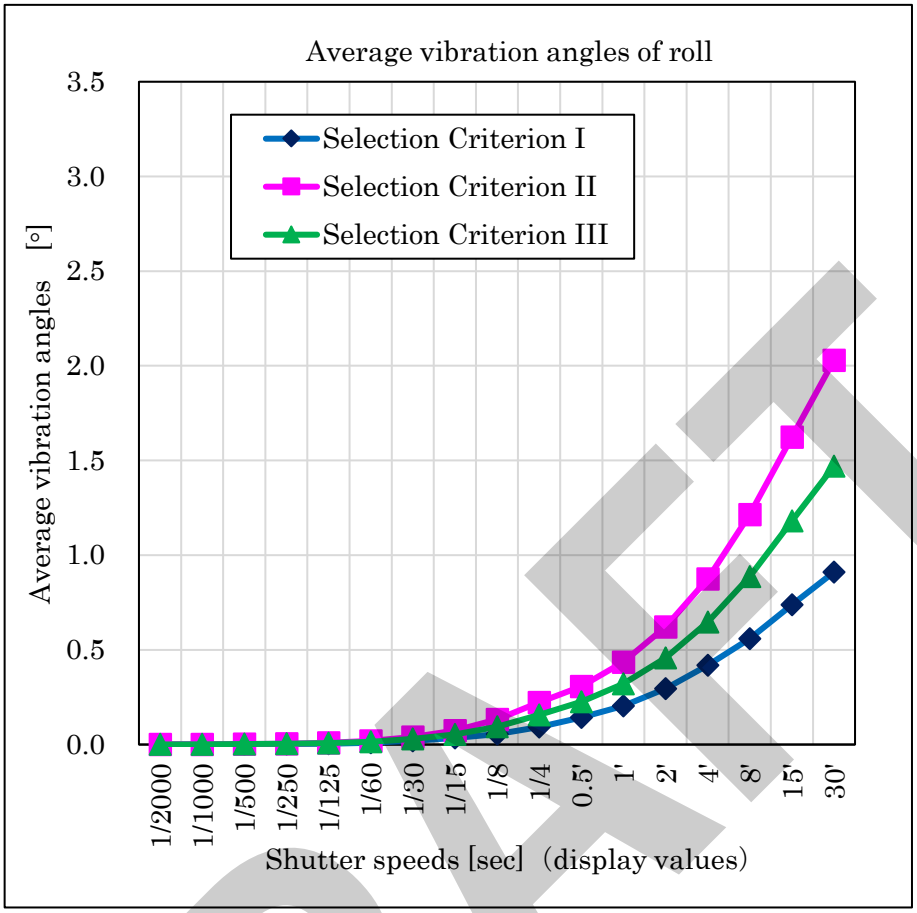


Figure 4-5-1d Average Vibration Angle of Roll

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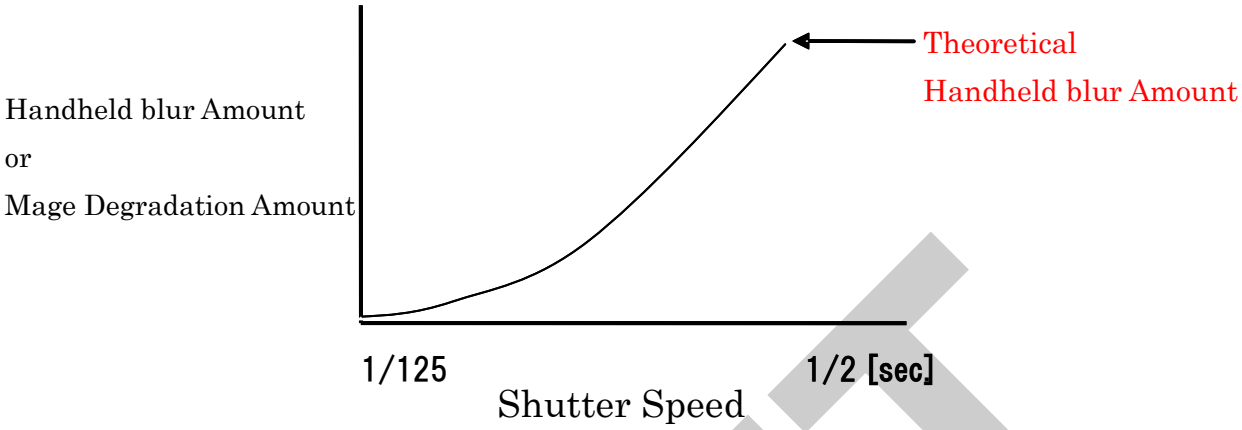


Figure 4-5-1e Theoretical Handheld Blur Amount

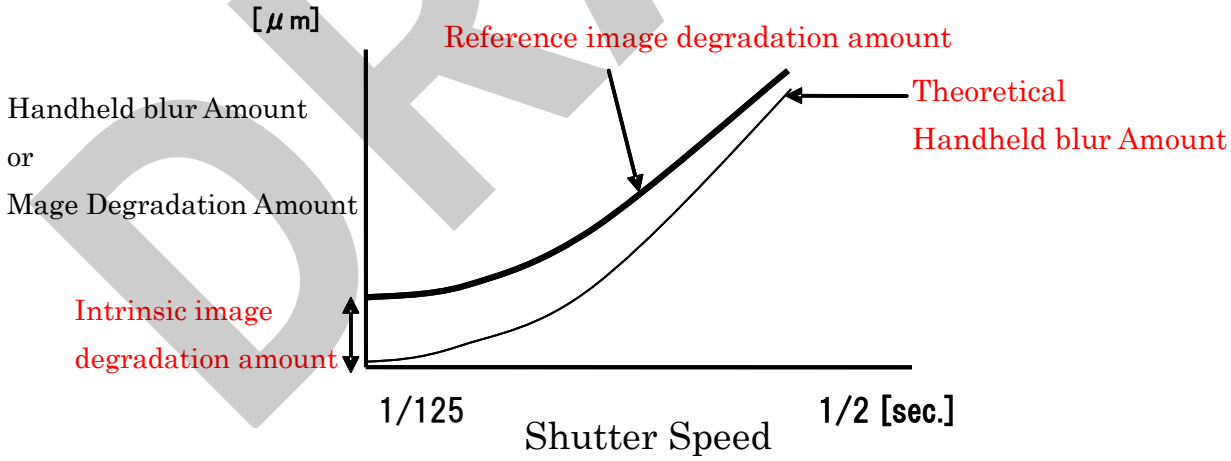


Figure 4-5-1f Reference image degradation amount

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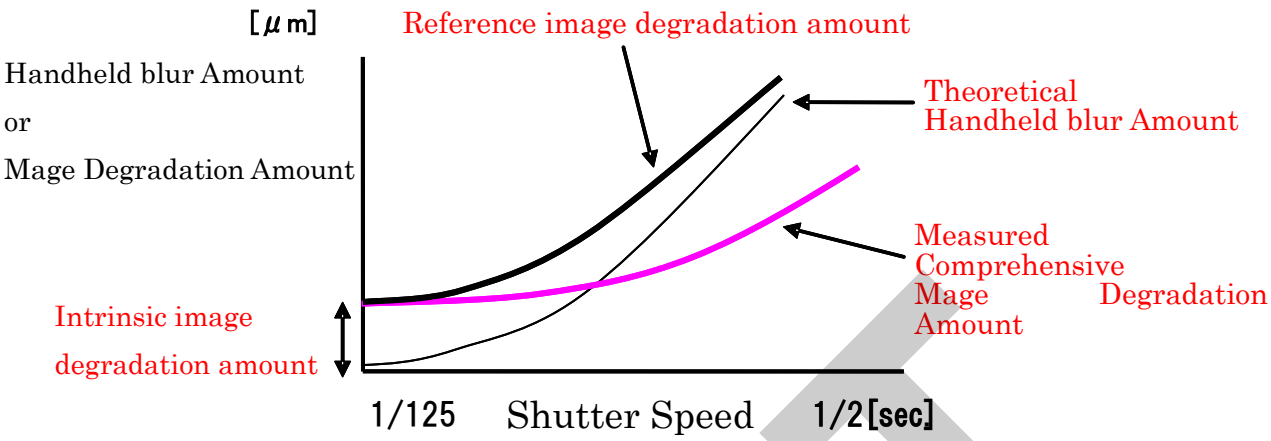


Figure 4-5-1g Measured image degradation amount

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4-5-2 Method of Converting Intrinsic Image Degradation Amount and Measured Image Degradation Amount into 35 mm Film Equivalent Values

If the units for intrinsic image degradation amount and total image degradation amount are left as the number of pixels, image stabilization performance cannot be determined. Thus, a method of conversion into 35 mm film equivalent values (the unit is μm) is described below.

Value on 35 mm film equivalent [μm] = Intrinsic Image Degradation Amount (or Total image degradation amount)[pixel] × (Diagonal Length of one Frame of 35 mm Film[μm]/ Diagonal Length of Taken Image[pixel])

1) Diagonal Length of one Frame of 35 mm Film

One frame of 35 mm film is 24 mm in the vertical direction, and 36 mm in the horizontal direction. Accordingly, the diagonal length can be calculated as follows:

$$\text{Diagonal Length[um]} = \sqrt{24^2 + 36^2} \times 1000$$

2) Diagonal Length of Image Taken

Diagonal Length of Image Taken [pixels] can be calculated as follows:

$$\text{Diagonal Length of Image Taken[pixel]} = \sqrt{(\text{the Number of Recorded Pixels in Vertical Direction})^2 + (\text{the Number of Recorded Pixels in Horizontal Direction})^2}$$

4-5-3 Image Stabilization Performance Calculation

Image stabilization performance is calculated on the basis of the reference handheld blur amount and measured handheld blur amount from the values calculated in 4-5-1. The image stabilization performance at the center shall be calculated using the reference handheld blur amount and the measured handheld blur amount at the center. If a performance description of image height at the corner of the image is required, the average value of the four positions at image height 60% shall be used as the image stabilization performance at the position of image height 60%. The average value of the four positions may be the number of stops obtained from the average of the measured handheld blur amounts at the four positions or the average of the number of stops obtained at each of the four positions. The number of stops at each of the four positions obtained in the process shall have a large

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enough number of decimal places to minimize the error when the number of stops is set to a 0.5 stop increment after averaging. In this method, difference between shutter speeds where both reference handheld blur amount and measured handheld blur amount are at the determination level for image stabilization performance is calculated as a stop number (0.5 stop increment and fractions less than 0.5 stop are dropped).

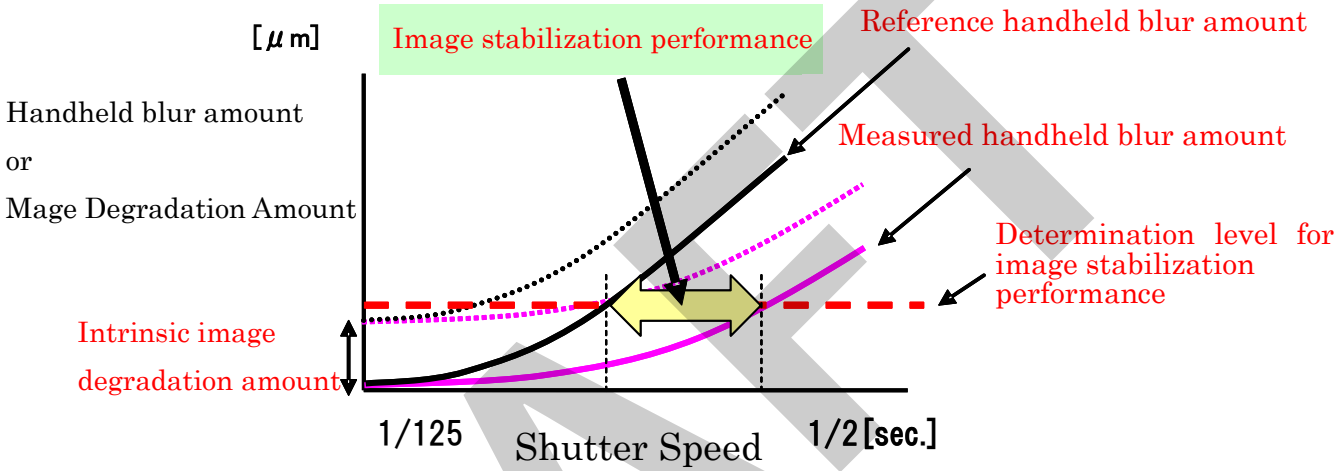


Figure 4-5-3a Image stabilization performance

Both reference handheld blur amount and measured handheld blur amount are discretely acquired at intervals of at most one stop. Accordingly, these amounts are calculated from the two shutter speeds on either side of the determination level for image stabilization performance.

Using Figure 4-5-3b as an example, SS_OFF, where the reference handheld blur amount becomes the determination level for image stabilization performance, is calculated by linearly interpolating the intersection on the logarithmic axis between the determination level for image stabilization performance and the line connecting reference handheld blur amount SS1 and reference handheld blur amount SS2.

Likewise, SS_ON, where the measured handheld blur amount becomes the determination level for image stabilization performance, is calculated by linearly interpolating the intersection on the logarithmic axis between the determination level for image stabilization performance intersection and the line connecting measured handheld blur amount SS3 and measured handheld blur amount SS4.

The difference between SS_OFF and SS_ON will give the number of stops of the image

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stabilization performance.

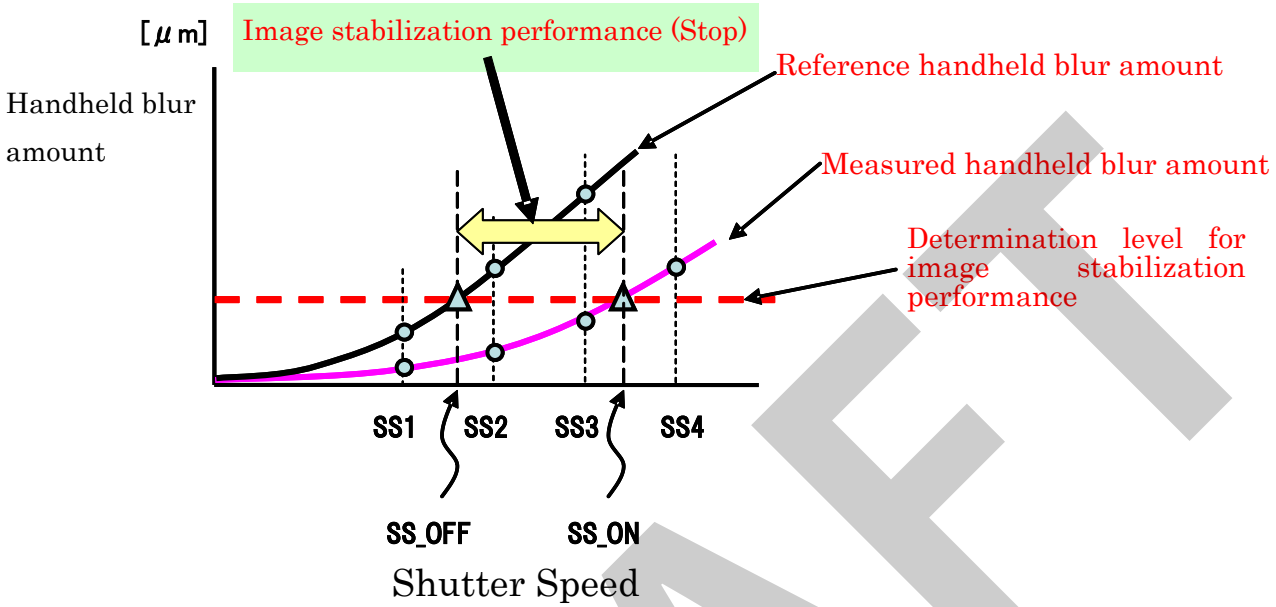


Figure 4-5-3b Image Stabilization Performance Calculation Method

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5. Description Method

5-1 Overview

This chapter specifies the description methods.

In the establishment of this standard, two vibration waveforms (WB-H and WB-L) are specified which quite faithfully simulate characteristics of camera shake as induced by handheld shooting. These two vibration waveforms were developed by analyzing extensive measurement data and adding further theoretical considerations.

In real situations, however, shooting generally starts by taking the camera out of a pocket or bag, adjusting certain settings, determining the shot composition by looking through a viewfinder or LCD monitor, and then pressing the shutter button. Thus, cameras tend to be subjected to large vibrations and shaking other than camera shake that arise during shooting. For some cameras, such factors may adversely affect the image stabilization performance at the instant of shooting which this standard is intended to address. Still, faithfully reproducing the adverse effect such vibrations have on measurements is not easy under this standard using a vibratory apparatus. It is also technically possible to produce a camera that is intentionally designed to generate favorable measurement results without regard to practicality with special settings that are adapted to the characteristics of the camera vibration waveforms specified in this standard.

Image stabilization performance in such cameras would be significantly lower than presented in the brochure or other documentation in actual handheld shooting. Accordingly, if the image stabilization performance for such cameras is presented in their brochures or the like as measured under this standard without modification, it would be misleading and detrimental to users. Therefore, suppliers shall take great care not to make any presentation of image stabilization performance that may be misleading or detrimental to users. If the image stabilization performance presented in the brochure or the like is only available under specific conditions, such conditions shall be expressly stated unless such conditions or facts are easily attainable by users from other information.

Performance and other specifications of the camera shall be measured according to the setup given in 4. Measurement Method in this standard and described based on the specifications in 5. Description Method. Here, description of measurement conditions and other information may be omitted except where mandatory.

Description of values measured using methods other than those in this standard is not prohibited, but such descriptions shall be made less prominently and measurement conditions shall be described.

If a numerical value to be described is an integer, its fractional portion may be omitted, irrespective of digits specified for each item. (Example: 2.0 stops ⇒ 2 stops)

Specific names, designations, Japanese long-sound symbols, and distinctions between single-byte and double-byte expressions may be used in customary expressions or names used by the suppliers or

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measurers provided that they do not cause misunderstanding. It shall be noted that:

Items clearly specified to be uniformly used and names defined in this standard shall follow such specifications; and names that may be confused with those defined in this standard shall not be used under a different definition.

The method of describing image stabilization performance in brochures shall be as follows.

Image stabilization performance shall be described using "stops." Values shall be rounded to the nearest 0.5-stop (e.g., 2.7 stops → 2.5 stops) and described as shown in the following example. Also, image stabilization performance may be described for multiple focal lengths. Here,

a) The factory default setting is recommended for image stabilization mode. When image stabilization performance is described by itself in this mode, the name of the image stabilization mode used in the measurement may be omitted.
b) When using an image stabilization mode other than the factory default setting, the name of the image stabilization mode used for measurement shall be clearly described.
c) For zoom lenses, the focal length (35 mm film equivalent) at which measurement is made or information replacing the focal length (such as “telephoto end”) shall clearly be described.
d) If the image stabilization performance presented is only available under specific conditions, such conditions shall be expressly stated unless such conditions or facts are easily attainable by users from other information.
e) When using the measurement methods of this standard to measure image stabilization for other than yaw, pitch and roll, take care not to mix yaw/pitch/roll corrections with corrections for other axes in the measurement results.
f) When denoting image stabilization for axes other than yaw, pitch and roll, either clearly state that the results were measured using proprietary measuring methods not conforming to CIPA standards, or add other expressions to that effect.
g) In specification tables or brochure descriptions, the presence/absence of a roll image stabilizer shall be clearly described.
h) The fact that measurements were made on the basis of this standard (CIPA Standard DC-X011–2024) shall be clearly indicated. If there is no risk of confusion with other CIPA standards, such an abbreviated description as CIPA2024 compliant is allowed to be used.

Note: Rules for description method of electronic image stabilization systems and of hybrid image stabilization systems that use both electronic and optical stabilization are specified in CIPA DCG-007 Description Method for Image Stabilization Performance of Digital Cameras (Electronic/Hybrid Systems).

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5-2 Performance Description of Lens Integrated Camera with Image Stabilization Mechanism

5-2-1 Lens Integrated Camera with Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance both at the center and at image height 60% shall be listed.

However, if 4-4-4 Action when Total Image Degradation Amount at Image Height 60% Is Not Measurable of 4 Measurement Method is applicable and the measurement at image height 60% is canceled, only the number of stops of the image stabilization performance at the center shall be presented; then an explanation indicating that measurement of the number of stops at image height 60% image height was not taken shall be provided.

In specification tables or brochure descriptions, the presence of a roll image stabilizer shall be clearly described.

Description Example 1

・Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 5-axis image stabilization, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:
Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization at a wide-angle end)
Center 6 stops and Corner 5.5 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization at a telephoto end), CIPA2024 compliant

Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 5-axis image stabilization, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops and Corner 4 stops* (yaw/pitch/roll performance: CIPA2024 compliant)
Reduced to 25% (shift performance: based on in-house measurement methods)
*At focal length of 50mm (35mm film equivalent: 100mm),
yaw/pitch/roll/shift image stabilization

Description Example 5: When measurement at corners canceled

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・Image stabilization performance: Center 5 stops, not measured at corner (yaw/pitch/roll performance, 5-axis image stabilization, IS auto of image stabilization mode at focal length of 600mm (35mm film equivalent: 900mm), CIPA2024 compliant)

5-2-2 Lens Integrated Camera with Image Stabilization in Yaw/Pitch Directions and without Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance at the center shall be described.

In specification tables or brochure descriptions, the absence of a roll image stabilizer shall be clearly indicated.

Description Example 1

・Image stabilization performance: Center 5 stops (yaw/pitch performance, 2-axis image stabilization, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:
Center 5 stops(yaw/pitch performance, yaw/pitch/shift image stabilization at focal length of 55mm (35mm film equivalent: 85mm))
Center 6 stops (yaw/pitch performance, yaw/pitch/shift image stabilization at focal length of 200mm (35mm film equivalent: 300mm)), CIPA2024 compliant

Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops (yaw/pitch performance, yaw/pitch/shift image stabilization, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops* (yaw/pitch performance: CIPA2024 compliant)
Reduced to 25% (shift performance: based on in-house measurement methods)
*At focal length of 50mm (35mm film equivalent: 100mm),
yaw/pitch/shift image stabilization

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5-3 Performance Description of Camera Body with Image Stabilization Mechanism

5-3-1 Camera Body with Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance both at the center and at image height 60% shall be listed.

However, if 4-4-4 Action when Total Image Degradation Amount at Image Height 60% Is Not Measurable of 4 Measurement Method is applicable and the measurement at image height 60% is canceled, only the number of stops of the image stabilization performance at the center shall be presented; then an explanation indicating that measurement of the number of stops at image height 60% image height was not taken shall be provided.

Lenses capable of utilizing the image stabilization performance shall be listed together by model name.

In specification tables or brochure descriptions, the presence of a roll image stabilizer shall be clearly described.

Description Example 1

・Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 5-axis image stabilization with AB lens, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:
Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization with AB lens at focal length of 55mm (35mm film equivalent: 85mm))
Center 6 stops and Corner 5.5 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization with AB lens at focal length of 200mm (35mm film equivalent: 300mm)), CIPA2024 compliant

Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 5-axis image stabilization with EF lens, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops and Corner 4 stops* (yaw/pitch/roll performance: CIPA2024 compliant)

Reduced to 25% (shift performance: based on in-house measurement methods)

*At focal length of 50mm (35mm film equivalent: 100mm),

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yaw/pitch/roll/shift image stabilization with GH lens

Description Example 5: When measurement at corners canceled

・Image stabilization performance: Center 5 stops, not measured at corner (yaw/pitch/roll performance, 5-axis image stabilization with AB lens, IS auto of image stabilization mode at focal length of 600mm (35mm film equivalent: 900mm), CIPA2024 compliant)

5-3-2 Camera Body with Image Stabilization in Yaw/Pitch Directions and without Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance at the center shall be described.

Lenses capable of utilizing the image stabilization performance shall be listed together by model name.

In specification tables or brochure descriptions, the absence of a roll image stabilizer shall be clearly indicated.

Description Example 1

・Image stabilization performance: Center 5 stops (yaw/pitch performance, 2-axis image stabilization with AB lens, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:
Center 5 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with AB lens at focal length of 55mm (35mm film equivalent: 85mm))
Center 6 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with AB lens at focal length of 200mm (35mm film equivalent: 300mm)), CIPA2024 compliant

Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with EF lens, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops* (yaw/pitch performance: CIPA2024 compliant)
Reduced to 25% (shift performance: based on in-house measurement methods)

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*At focal length of 50mm (35mm film equivalent: 100mm),
yaw/pitch/shift image stabilization with GH lens

5-4 Performance Description of Interchangeable Lens with Image Stabilization Mechanism

5-4-1 Interchangeable Lens with Image Stabilization in Yaw/Pitch Directions and without Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance at the center shall be described.

Camera bodies capable of utilizing the image stabilization performance shall be listed together by model name.

Camera body used for measurement may be taken with or without image stabilization. However, if possible, the image stabilization of the camera body used for measurement be disabled.

In specification tables or brochure descriptions, the absence of a roll image stabilizer in the lens shall be clearly stated.

Description Example 1

・Image stabilization performance: Center 5 stops (yaw/pitch performance, lens with 2-axis image stabilization, IS auto of image stabilization mode with AB camera at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:
Center 5 stops (yaw/pitch performance, lens: yaw/pitch image stabilization with AB camera at a wide angle end)
Center 6 stops (yaw/pitch performance, lens: yaw/pitch image stabilization with AB camera at telephoto end), CIPA2024 compliant

Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops (yaw/pitch performance, lens: yaw/pitch/shift image stabilization with all cameras, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops* (yaw/pitch performance: CIPA2024 compliant)
Reduced to 25% (shift performance: based on in-house measurement methods)

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*At focal length of 50mm (35mm film equivalent: 100mm),
lens: yaw/pitch/shift image stabilization with AB camera

5-5 Performance Description for Pairing a Camera Body with Image Stabilization Mechanism and an Interchangeable Lens with Image Stabilization Mechanism

5-5-1 Pairing with Image Stabilization in Yaw/Pitch/Roll Directions

Provisions

The number of stops of image stabilization performance both at the center and at image height 60% shall be listed.

However, if 4-4-4 Action when Total Image Degradation Amount at Image Height 60% Is Not Measurable of 4 Measurement Method is applicable and the measurement at image height 60% is canceled, only the number of stops of the image stabilization performance at the center shall be presented; then an explanation indicating that measurement of the number of stops at image height 60% image height was not taken shall be provided.

The camera body and lens capable of achieving the image stabilization performance shall be listed together by model name.

In specification tables or brochure descriptions, the presence of a roll image stabilizer shall be clearly described.

Description Example 1

- Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 5-axis image stabilization with AB camera and AB lens, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

- Image stabilization performance:
  Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization with AB camera and AB lens at focal length of 55mm (35mm film equivalent: 85mm))
  Center 6 stops and Corner 5.5 stops (yaw/pitch/roll performance, yaw/pitch/roll/shift image stabilization with AB camera and AB lens at focal length of 200mm (35mm film equivalent: 300mm)), CIPA2024 compliant

Description Example 3: Prime lenses only

- Image stabilization performance: Center 5 stops and Corner 4 stops (yaw/pitch/roll

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performance, 5-axis image stabilization with EF camera and EF lens, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops and Corner 4 stops* (yaw/pitch/roll performance: CIPA2024 compliant)

Reduced to 25% (shift performance: based on in-house measurement methods)

*At focal length of 50mm (35mm film equivalent: 100mm),
yaw/pitch/roll/shift image stabilization with GH camera and GH lens

Description Example 5: When measurement at corners canceled

・Image stabilization performance: Center 5 stops, not measured at corner (yaw/pitch/roll performance, 5-axis image stabilization with AB camera and AB lens, IS auto of image stabilization mode at focal length of 600mm (35mm film equivalent: 900mm), CIPA2024 compliant)

5-5-2 Pairing with Image Stabilization in Yaw/Pitch Directions and without Image Stabilization in Roll Direction

Provisions

The number of stops of image stabilization performance at the center shall be described.

The camera body and lens capable of achieving the image stabilization performance shall be listed together by model name.

In specification tables or brochure descriptions, the absence of a roll image stabilizer shall be clearly indicated.

Description Example 1

・Image stabilization performance: Center 5 stops (yaw/pitch performance, 2-axis image stabilization with AB camera and AB lens, IS auto of image stabilization mode at focal length of 55mm (35mm film equivalent: 85mm), CIPA2024 compliant)

Description Example 2

・Image stabilization performance:

Center 5 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with AB camera and AB lens at focal length of 55mm (35mm film equivalent: 85mm))

Center 6 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with AB camera and AB lens at focal length of 200mm (35mm film equivalent: 300mm)), CIPA2024 compliant

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Description Example 3: Prime lenses only

・Image stabilization performance: Center 5 stops (yaw/pitch performance, yaw/pitch/shift image stabilization with EF camera and EF lens, CIPA2024 compliant)

Description Example 4: Cameras with image stabilization for other than just yaw/pitch/roll

・Image stabilization performance: Center 5 stops* (yaw/pitch performance: CIPA2024 compliant)

Reduced to 25% (shift performance: based on in-house measurement methods)

*At focal length of 50mm (35mm film equivalent: 100mm),
yaw/pitch/shift image stabilization with GH camera and GH lens

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6. Explanation

6-1 Background of Specifying Measurement Method

This chapter specifies measurement methods in an effort to contribute to sound development within the industry through the realization of fair competition.

When measuring image stabilization performance, the most understandable way is to compare the results with image stabilization OFF against those with image stabilization ON and quantitatively describe the difference as the effect of image stabilization. When doing so, the intrinsic image degradation amount is first removed from the figures for both the ON and OFF states before comparison. This allows for image degradation elements caused by factors other than camera shake to be easily removed and produces fairer measurement results.

Meanwhile, in the measurement method for this standard, the camera to be measured is vibrated using a vibratory apparatus with the vibration waveforms specified by the standard and prescribed performance. Accordingly, handheld blur amounts with image stabilization OFF should theoretically be similar among cameras, exhibiting only subtle individual differences. Results of measurements for this standard performed on many cameras have confirmed that in actuality, there is no significant difference among cameras.

With fully automated cameras already on the market, there are already cameras without a mode where image stabilization is OFF. This makes it difficult to measure all the cameras based on whether image stabilization is ON or OFF. This trend is predicted to grow even further in the future.

Considering the errors that occur with measurement and the man-hours involved, completing measurement with image stabilization ON only instead of measuring twice with image stabilization both ON and OFF will cut the chances for measurement errors in half and also reduce the work load.

For the above reasons, in this standard, image stabilization performance is calculated from the reference handheld blur amount, which corresponds to measurement results with image stabilization OFF, and measured handheld blur amount, which is the measurement result with image stabilization ON.

6-2 Vibratory Apparatus

In measuring a camera’s image stabilization performance, the most realistic scenario is to shoot the handheld blur measurement chart with the camera by hand. The scale and characteristics of camera shake will vary among different individuals, however, and the same person may even produce different results with a different camera or shooting conditions.

To specify the methods for measurement and describe image stabilization performance, such uncertainties shall be eliminated. Using a mechanical device to generate vibration simulating camera

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shake and transmitting the vibration to the camera is an effective means of achieving this affect. Such devices are called vibratory apparatuses. A vibratory apparatus is designed to reproduce camera shake by human hands with an electric signal called a vibration waveform applied to vibrate the movable part of the apparatus.

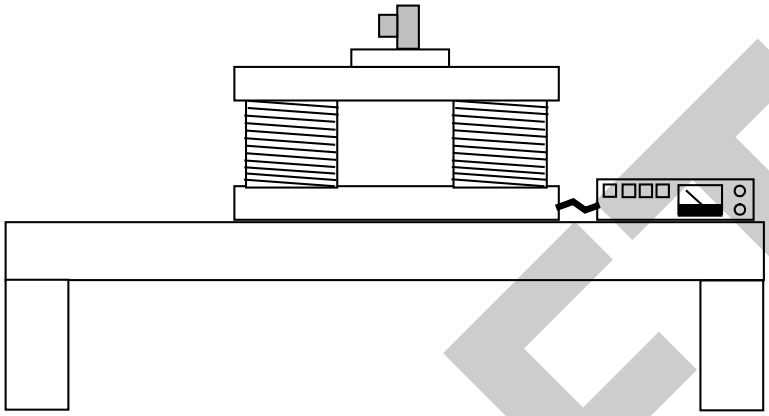


Figure 6-2-1 Vibratory apparatus

6-3 Vibration Waveform

Preferably, the vibration waveform will be as close as possible to camera shake characteristics. Accordingly, CIPA has collected camera shake frequencies and amplitudes from many people by attaching a sensor to a camera and separating them into yaw, pitch and roll. Vibration waveforms were generated by analyzing the collected data and extracting characteristic frequency and amplitude to synthesize the waveforms.

Handheld blur has six directional components in total: yaw, pitch, roll, X, Y, and Z. At a shooting distances of about 20 times the focal length, however, yaw and pitch followed by roll are dominant, making the other directions practically negligible. Thus, waveforms with only three-axis components were adopted.

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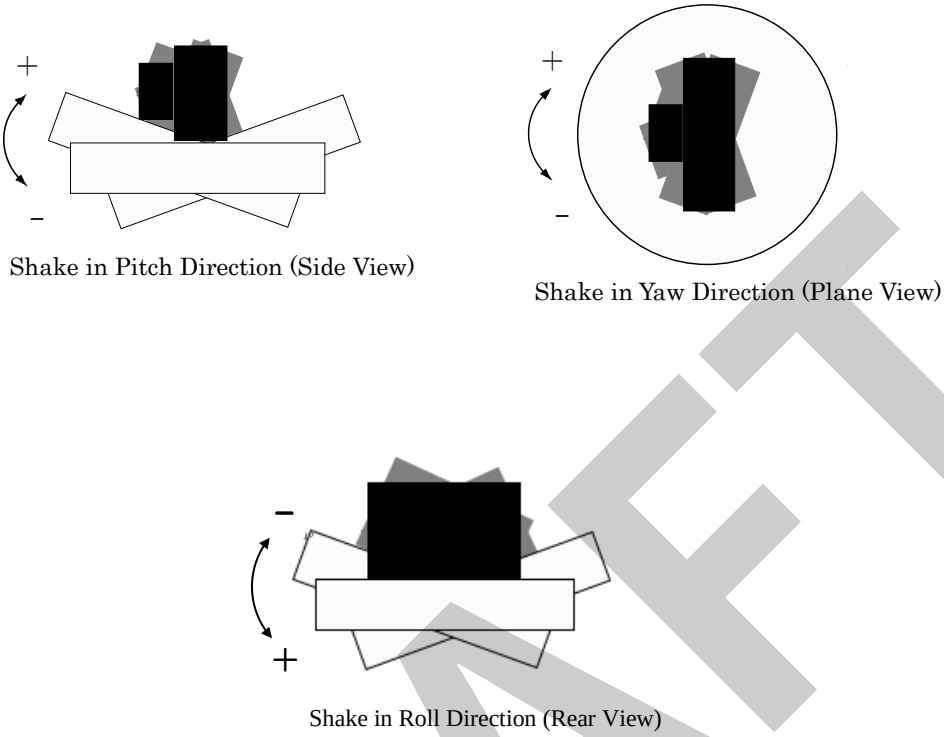


Figure 6-3a Shake Directions

In the process of analyzing the characteristics of camera shake, the frequency and amplitude of the camera shake have been verified as differing between interchangeable lens cameras and compact cameras. One major factor is the difference in total mass. Accordingly, two types of vibration waveforms have been created to include optimal amplitude and frequency components for both camera segments: WB-H for interchangeable lens cameras and WB-L for compact cameras.

In principle, the two waveforms are used for different purposes in carrying out measurements: one for interchangeable lens cameras and the other for compact cameras. However, there are some compact cameras with total mass close to that of interchangeable lens cameras, and further a new category of cameras called non-reflex (mirrorless) cameras have now hit the market. For cameras with a total mass somewhere between that of a compact camera and interchangeable lens camera, using only one of the two waveform types has been confirmed to produce a discrepancy between the measured image stabilization performance and image stabilization performance* as acquired in a shooting experiment with many test subjects (Figure 6-3b).

Thus, the type of vibration waveform to use for calculating image stabilization performance is determined by total mass instead of camera category. WB-L is used for total masses of less than 400

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g, WB-H is used for total masses of 600 g and above, and both of the vibration waveforms are used for cameras between 400 g and 600 g. This is a result of statistical calculation based on results of measurement on many cameras.

Taking the weighted average of measurement results according to the total mass when both types of vibration waveforms are used was originally considered, but there was no significant difference in results despite the more complex computation. Thus, a standard arithmetic method was settled upon.

* In the experiment, image stabilization performance was measured from images taken with image stabilization ON and with image stabilization OFF. Image stabilization performance was measured according to the method described in 4-4. Performance was calculated by comparing values corresponding to the measured handheld blur amount acquired from images taken with image stabilization ON and with image stabilization OFF at the determination level for image stabilization performance. In the experiment, however, yaw and pitch waveforms were used to calculate the image stabilization performance at the center of the image, and not the roll waveform.

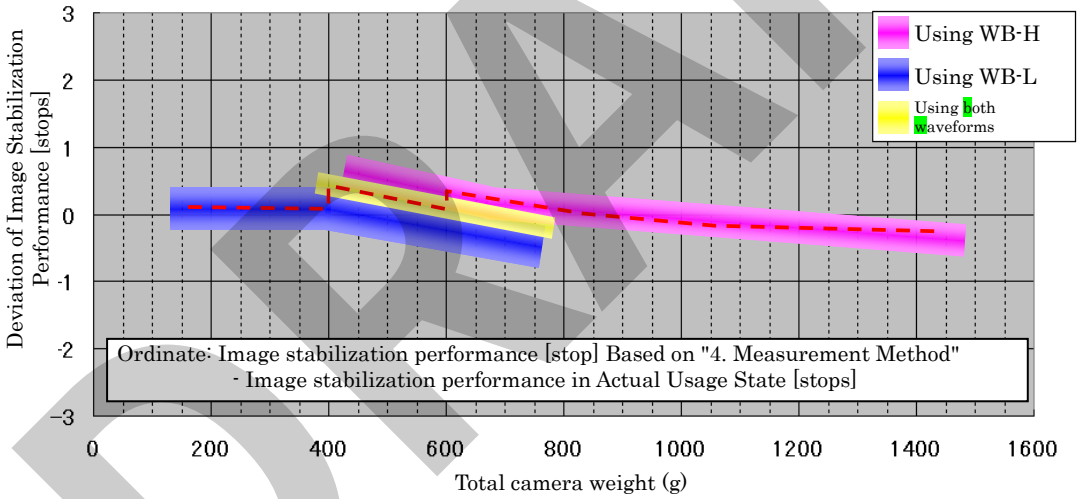


Fig. 6-3b Test Result

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6-4 CIPA Handheld Blur Measurement Chart

To facilitate focusing and for general versatility, the CIPA handheld blur measurement chart uses natural images as shown in Figure 4-2-1.

6-5 Handheld Blur Measurement Software

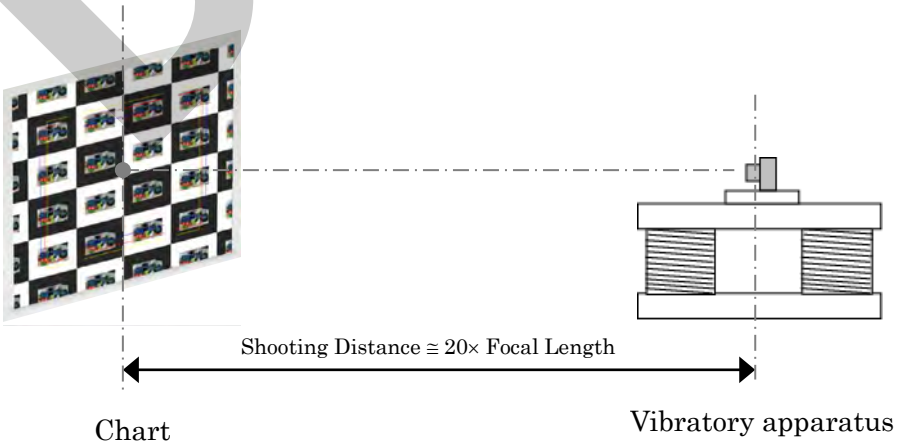
In finding the intrinsic image degradation amount and total image degradation amount, while it is possible, for example, to visually read out the width of the boundary region between the black and white portions of the chart, it is not possible to eliminate variations among measurements. Thus, handheld blur measurement software was developed to eliminate variation and for improved convenience.

6-6 Shooting Distance

In principle, the distance between the handheld blur measurement chart and the camera to be measured when measuring image stabilization performance shall be about 20 times the 35 mm film equivalent focal length. This distance has been calculated in light of practical aspects to minimize doubt about measurement results and the experiment environment. As an example, the shooting distance for a camera with a 35 mm film equivalent focal length of 100 mm is about 2 m.

For a wide-angle lens having a 35 mm film equivalent focal length of 24 mm, the shooting distance will be no more than 500 mm. In this situation, the handheld blur measurement chart, vibratory apparatus, and lighting device can interfere, making measurement difficult. Conversely, the shooting distance for a telephoto lens having a 35 mm film equivalent focal length of at least 500 mm would be 10 m or more, which could possibly make a shooting site with the proper conditions impractical.

In such unavoidable cases, shooting distance may be increased or shortened from 20 times the 35 mm film equivalent focal length by the minimum amount required.



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Figure 6-6 Shooting Distance

6-7 Number of Shots

The vibration waveforms shown in 6-3 are used to measure image stabilization performance in this standard. However, there may be image stabilization performance errors due to bias in the data collected if not enough shots are taken.

Thus, the specified minimum number of shots for each shutter speed in this standard is 200. Shooting experiments (shooting at a shutter speed of 1/8 second) on many test subjects have statistically revealed that taking roughly 200 images can suppress 95% of the variations in every measurement of total image degradation amount to the determination level for image stabilization performance or less.

6-8 Determination Level for Image Stabilization

With handheld blur amount from images being the variable measured to determine the image stabilization performance, it can be challenging to measure performance when there is only a little image degradation or handheld blur.

In photography, the diameter of what is called the circle of confusion is often used for determining image degradation. The largest circle of confusion incapable of determining image degradation and handheld blur is referred to as the permissible circle of confusion. The size of the permissible circle of confusion will differ based on factors such as the size of the picture viewed, viewing distance and the basis for calculation.

Thus, a level was derived by experiment that is close to the size of the permissible circle of confusion and leaves no doubts about the performance measurement that is the determination level for image stabilization performance.

In recent years, the viewing environment of photographs has become more diverse, no longer being limited to postcard size, and includes A4 or larger prints as well as viewing by computer monitors, smartphones, tablets, and television monitors. With such diversity, photographs and images to be viewed are expanding more than before.

Specifically, images were created by enlarging the diameter of the circle of confusion incrementally by computer simulation, and then prints of various sizes of A4 and larger were subjected to panel testing at viewing distances from which the entire image could be seen. Panel tests were conducted using a 24-inch PC monitor as a representative of self-illuminating displays, with the entire image displayed at a normal viewing distance.

Generally, sensory evaluations of image quality will have a reference set at a point where at least 10% of panelists can recognize the difference in image quality. In the two panel test, approximately

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10% of panelists could determine image degradation or blur in the permissible circle of confusion at a diameter of about 24 μm.

The permissible circle of confusion for each viewing device was calculated on the basis of the screen size and viewing distance. That is, given 1 minute as the angle between two points which a person with an eyesight of 1.0 (20/20 vision) is capable of discriminating and 600mm as the viewing distance, for example, then, on the viewing device screen, this is equivalent to:

tan (1/60°) × 600 [mm] × 1000 [μm/mm] = 175 [μm]

Therefore, image degradation of 175 μm or less on the viewing device display screen can be said to be almost unrecognizable by the viewer.

If this viewing device is an A4 print (diagonal length 364mm) and the camera is a 35mm full size (diagonal length 43.3mm) camera, then, on the imaging element,

175 [μm] × 43.3 [mm] / 364 [mm] = 21 [μm]

is the equivalent image degradation amount.

From the above result, the determination level for image stabilization performance was set to 20 μm.

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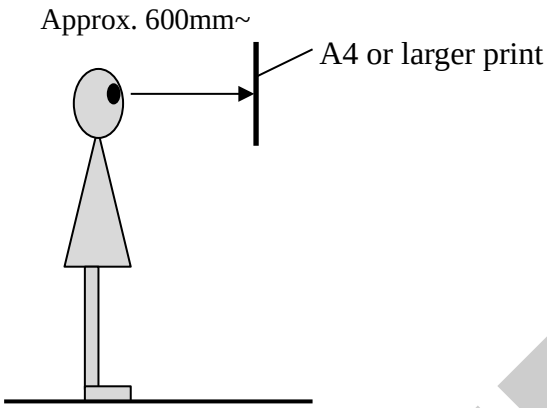


Figure 6-8a Print Viewing Example

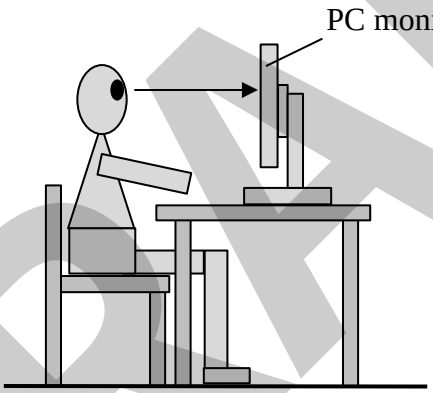


Figure 6-8b PC Monitor Viewing Example

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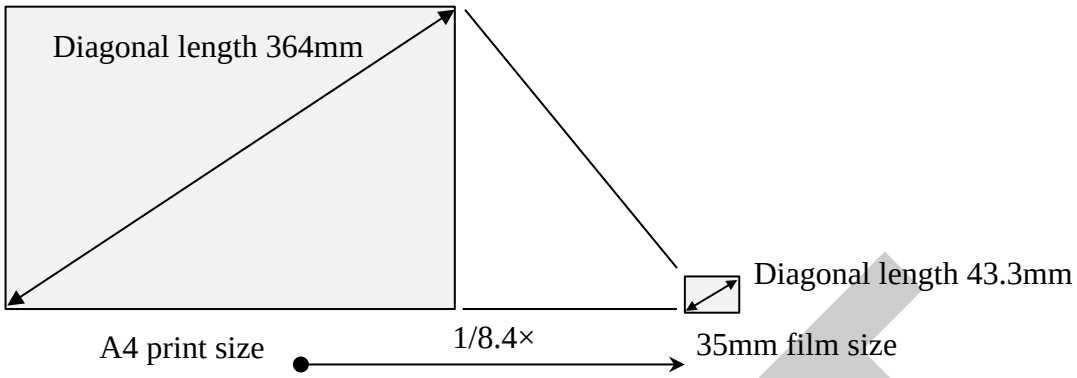


Figure 6-8c A4 Size and 35mm Film

6-9 Image Stabilization Performance Calculation

Generally, the measured handheld blur amount rapidly increases as shutter speeds slow down. Accordingly, the actual handheld blur amount is a curve connecting handheld blur amounts acquired discretely at intervals of at most one stop. While deriving the precise function for this curve would require advanced mathematical techniques, linear approximation does not produce significant error as long as intervals are at most one stop. Further, as the reference handheld blur amount forms a curve similar to that of the measured handheld blur amount, linear approximation of these is expected to have the errors in both cancel each other out and further reduce error.

Thus, this standard avoids using a complex curve for calculating image stabilization performance with linear approximation connecting two points with a straight line.

6-10 Lighting

Since the type and color temperature of lighting are unlikely to greatly affect measurement results, no type of lighting is specified. However, the full handheld blur measurement chart shall be evenly lit during measurement. Accordingly, the number of lights, positioning and other layout details will require consideration.

Brightness of lighting is to be adjustable in case brightness adjustments are required to change the shutter speed for settings in which shutter speed cannot be changed easily by camera operation. In such cases, combined use with a neutral density (ND) filter can be effective.

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6-11 Shooting Mode

With regard to the shooting mode for measuring image stabilization performance, while no specific shooting mode is stipulated, shooting modes will preferably be capable of remembering as many of the items specified in 4-3-3 to 4-3-19 as possible and setting them in advance with the power OFF given that shooting shall begin within three seconds after shooting is enabled following power ON. This is required because shooting cannot start immediately if camera settings for measurement are adjusted after the power is turned ON.

Next, the option of a mode "allowing for easy changing of the shutter speed" is mentioned. This is recommended to make it easier to change shutter speeds when fulfilling the requirement to measure handheld blur amounts at each shutter speed.

Many cameras, however, only have shooting modes such as "Auto" and "Program" that automatically change shutter speed and aperture values according to the shooting environment and state of the subject. For such cameras, measures will be required for changing the shutter speed, such as adjusting the brightness of lighting.

6-12 Image Quality Mode (Compression Ratio)

While no compression ratio is specified, image compression may result in noise effects which raise doubts about the measurement results. Accordingly, a high image quality mode with low compression ratio is preferable.

6-13 Image Quality Mode (Number of Recording Pixels)

While no number of recording pixels is specified, fewer recording pixels may degrade resolution to the point that doubts are raised about the measurement results. Accordingly, taking images at the maximum number of pixels offered by the camera as not to impact resolution is preferable.

6-14 ISO Sensitivity

While no ISO sensitivity is specified, the noise effects from shooting at high sensitivity may raise doubts about the measurement results. Accordingly, ISO sensitivity should be set relatively low to prevent noise.

However, some cameras do not allow the user to set ISO sensitivity. For such cameras, make every effort not to render questionable measurement results. As noise levels change with ISO sensitivity changes, camera settings should prevent ISO sensitivity changes where possible.

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6-15 Flash

Flashes may change image quality by emitting light and raise doubts about the measurement results. Many cameras do not allow shutter speed adjustments with the flash activated. Accordingly, measurement shall be made with the flash OFF.

6-16 Storage Media

Since types and capacities of storage media are not likely to raise issues with measurement results, none are specified. However, storage media of extremely small capacity will have to be replaced frequently during measurement and will require data to be deleted. This would be a concern as changing camera settings may result in inconsistent measurements.

Accordingly, storage media with sufficiently large capacity, that will not have to be replaced during measurements, are more convenient.

6-17 Focal Length for Measurements (Cameras Mounted with Zoom Lenses)

With optical zoom factors and focal length ranges varying from product to product, not every digital camera can be measured at the same focal length. At first, the possibility of methods of selecting from several predetermined focal lengths at both the wide-angle and telephoto end was considered. Given that some cameras cannot select specific focal lengths, the decision on a focal length for measurement will be left up to suppliers and measurers. Meanwhile, some cameras also offer different image stabilization performance at different focal lengths. For this reason, the focal length for the measurement shall be clearly labeled when listing image stabilization performance.

6-18 Electronic (Digital) Zoom

Electronic (digital) zoom shall not be used as it deteriorates image quality, which may raise doubts about measurement results.

6-19 Focus Control

Whether autofocus (AF) or manual focus (MF) is used, measurements cannot distinguish between handheld blur and the lens being out of focus if the camera is not properly focused on the handheld blur measurement chart. This could possibly raise questions about measurement results.

Accordingly, the camera shall be properly focused on the handheld blur measurement chart when

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measuring.

6-20 White Balance

While white balance settings are not specified, shooting with extremely unbalanced color temperature settings could raise questions about measurement results. Accordingly, white balance is to be set appropriately for the light source to be used.

6-21 Image Aspect Ratio

While some cameras are capable of changing the image aspect ratio during imaging, differences in image aspect ratio are unlikely to raise issues with measurement results for image stabilization performance. There is thus no need to specify the image aspect ratio. The factory default setting mode is recommended.

6-22 Image Stabilization Mode

Cameras may have one of several types of image stabilization modes. Some modes keep image stabilization constantly on, while in others image stabilization may only be turned on when the shutter is released. Some even have variable image stabilization characteristics depending on the shooting mode and zoom position.

Generally, most cameras will have a factory default auto shooting mode. As suppliers recommend shooting mode to be set to auto, image stabilization modes where the camera operates in this state should be used.

In some cases, settings for image stabilization are not available in the factory default shooting mode, or image stabilization modes other than the factory default setting may exhibit higher image stabilization performance at a certain focal length. In these cases, the supplier or measurer is not precluded from selecting the appropriate image stabilization mode. However, if an image stabilization mode other than the factory default setting is used, the name of the image stabilization mode used in measurement shall be clearly listed.

6-23 Necessity of Measurement at Image Height 60%

If hand-held night shots are taken with a camera that corrects only for yaw and pitch handheld blur, there may still be blur-like image degradation at the corner of the image. In recent years, the range in which AF (auto focus) is possible within an image has expanded, and the main subject is often placed

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in the corner of the screen.

Therefore, in addition to measurement at the center of the image, the decision was made to add measurement at positions of image height 60% at the corners of the image.

Camera Shake Characteristics

Camera shake has components in six axes total (yaw, pitch, roll, X, Y, and Z). For the sake of explanation, assume that there is a left-handed coordinate system in three-dimensional space (Cartesian coordinate system with X, Y, and Z axes) and that the optical axis of the imaging lens is coincident with the Z axis.

The X, Y, and Z components are the handheld blur in which reciprocating translational motion occurs in the X-, Y-, and Z-axis directions, respectively. The Z component is the handheld blur in the optical axis direction. The remaining X and Y components are the handheld blur in planes perpendicular to the optical axis. The vertical and horizontal handheld blur together are generally referred to as the shift component or shift handheld blur.

The yaw, pitch, and roll components are the handheld blur in which reciprocating rotational motion occurs about the X axis, Y axis, and Z axis, respectively. The reciprocating rotational motion about the Y axis and the reciprocating rotational motion about the X axis are called the yaw component and the pitch component. The reciprocating rotational motion about the optical axis (Z axis) is called the roll component.

The X, Y, and Z components, when converted to the amounts of handheld blur on an imaging surface, are proportional to the magnification of the image. When the shooting distance is 20 times the focal length or more, the amount of handheld blur decreases considerably, and the yaw, pitch, and then roll components become dominant in the amount of handheld blur on the imaging plane, and the other three axis components of X, Y, and Z can be virtually ignored.

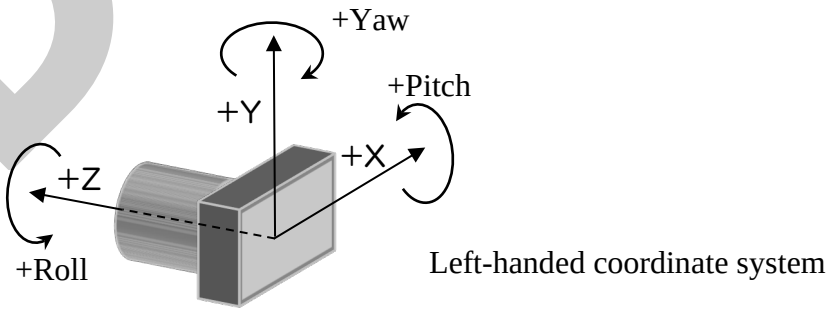


Figure 6-23a Coordinate System

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Degree of Image Height Impact on Roll Component

The roll component is a reciprocating rotational handheld blur about the optical axis (Z axis), resulting in a Handheld Blur amount of zero at the center of the image plane (image height 0%). The handheld blur amount then increases in proportion to the image height.

In this standard, the vibration waveforms are those of the three axis components of yaw, pitch, and roll. While the number of stops of the image stabilization performance for the yaw and pitch components can be measured at the center of the image plane (image height 0%), measurement of the number of stops of the image stabilization performance for yaw, pitch, and roll components requires measurement at a certain image height at the corners of the image.

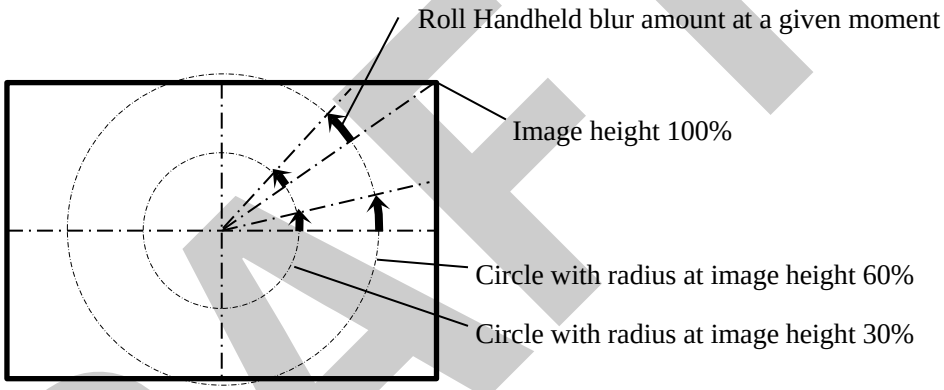


Figure 6-23b Roll Handheld blur Amount at Given Moment

Reason for Measuring Corner of Image at Height 60%

When the positions of main subjects (people, eyes, animals, plants, etc.) were measured in multiple photographs, more than 95% of the subjects were positioned within a radius of image height 60% about the center of the imaging plane.

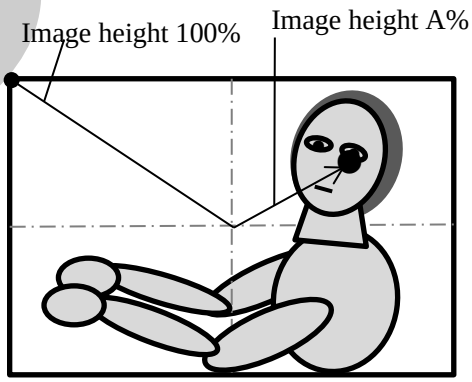


Figure 6-23c Example of Main Subject Position (Eye Position)

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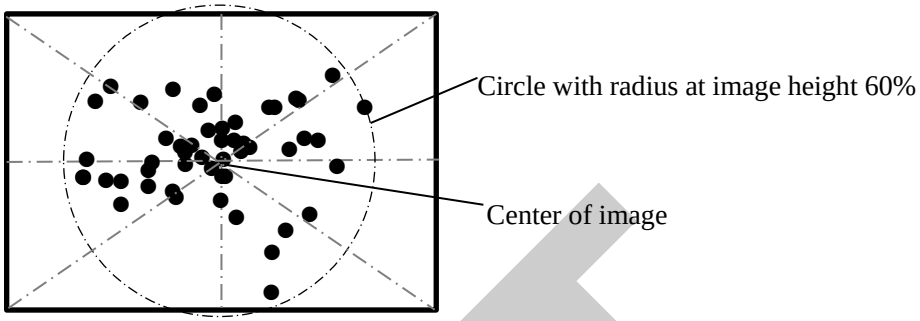


Figure 6-23d Distribution of Main Subject Positions

The decision was therefore made to measure the number of stops of image stabilization performance at two positions: the center of the imaging plane, where many main subjects are present, and at image height 60%, which is an outer edge where main subjects are present.

Reason for Use of Average Value of Number of Stops at Four Corners

For the measurement at image height 60%, a total of four locations at points B in Figure 4-23e on diagonal lines are measured, and an average value of the number of stops of the image stabilization performance is used.

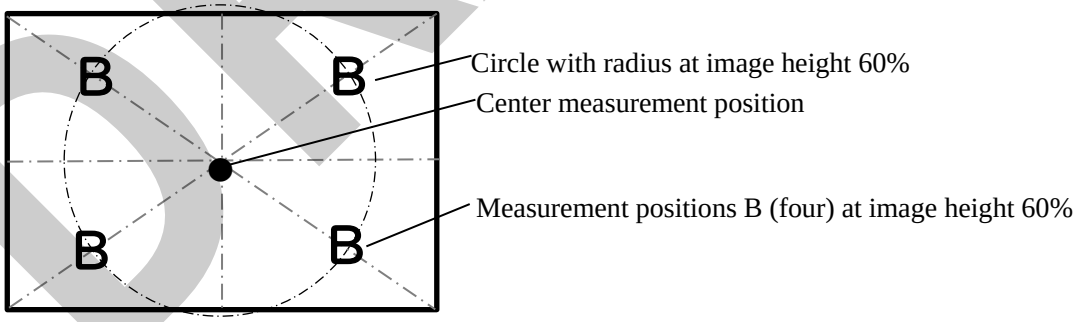


Figure 6-23e Measurement Positions of Number of Stops of Image Stabilization Performance

When the remaining handheld blur components following camera based image stabilization is checked, the center of rotation of the roll residual handheld blur is sometimes shifted from the center of the image. As a result, the number of stops of image stabilization performance at the four positions

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on the diagonal lines at image height 60% may differ in value. To mitigate this variation, the average value of the four locations is used.

As illustrated in the figure below, if the center of rotation of the roll handheld blur of the remaining handheld blur is shifted from the center of the image during roll image stabilization, the remaining roll handheld blur amounts at the four B positions differ, increasing as the distance from the center of rotation of the roll handheld blur of the remaining handheld blur increases. To indicate performance in detail, it would be best to indicate the number of stops of image stabilization performance at the four locations; however, because doing so would result in an increase in number of display values, the decision was made that the average value of the four positions at image height 60% shall be presented.

This average value may be the number of stops obtained from the average of the measured handheld blur amounts at the four positions or may be the average of the number of stops obtained at each of the four positions.

If the number of decimal places in the number of stops at each of the four positions obtained in the process has large enough to minimize the error when the number of stops is set to a 0.5 stop increment after averaging, the difference between these two averaging methods becomes minute.

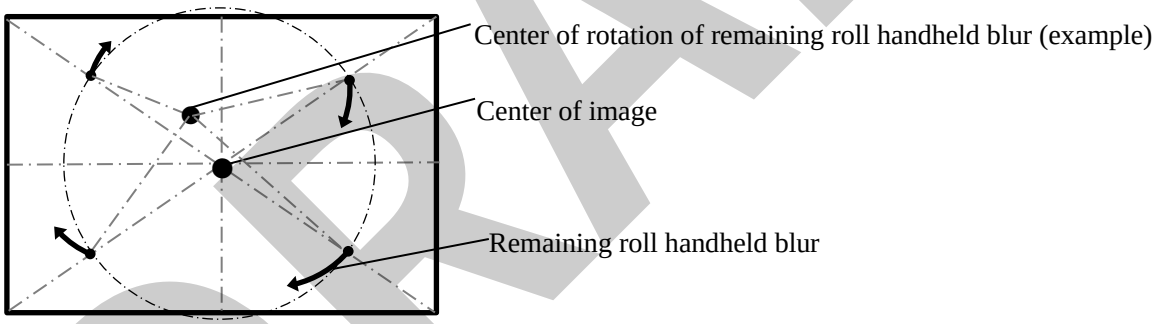


Figure 6-23f Example of Shifted Center of Rotation of Remaining Roll Handheld Blur

6-24 Theoretical Handheld Blur Amount when Yaw/Pitch/Roll Handheld Blur Present

As described in 2) Theoretical Handheld Blur Amount of 4-5-1 Calculation of Basic Values Required to Calculate Image Stabilization Performance, the following equality is defined in this standard:

$$\text{Theoretical handheld blur amount [μm]} = \sqrt{(\text{Yaw/pitch theoretical handheld blur amounts [μm]})^2 + (\text{Roll theoretical handheld blur amount [μm]})^2}$$

The yaw and pitch vibration waveforms and roll vibration waveform excited by the vibration table are vectors in the same plane when observed on the imaging plane of a camera and, when considered

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in the context of a single image taken at a certain shutter speed, have components that add to and cancel out each other. Therefore, to strictly obtain a theoretical handheld blur amount, preferably simulation is performed for each focal length and each shutter speed using the vibration waveforms.

However, the correlation between the yaw/pitch vibration waveforms and the roll vibration waveform is low and, even if the theoretical handheld blur amount is statistically calculated as the square root of the sum of the squares of the yaw/pitch theoretical handheld blur amount and the roll theoretical handheld blur amount, the difference from the value strictly obtained from such simulation is small.

Therefore, in this standard, the theoretical handheld blur amount is determined by the square root of the sum of the squares. The advantage of this method is, as long as there are two tables of numerical values, namely the yaw/pitch average vibration angle and the roll average vibration angle for each shutter speed, the simplicity of calculating the theoretical handheld blur amount at any focal length using a mathematical formula.

On the other hand, the method of strictly obtaining the theoretical handheld blur amount by simulation requires the measurer to obtain the theoretical handheld blur amount for each shutter speed at all focal lengths from wide-angle to telephoto.

6-25 Description Method

Meaning of Each Section of Performance Description

In this revision, the main section of the performance description example for, for example, a camera body with a yaw/pitch/roll image stabilization mechanism, is A stops at center, B stops at corner (yaw/pitch/roll performance, *).

The words "center" and "corner" written before the number of stops are easy words to indicate the location of the performance measurement within the screen. Center means "performance at the center position of the image at image height 0%" and corner means "performance at a corner position at image height 60%."

The “yaw/pitch/roll performance” written after “Center A stops and Corner B stops” means that the comprehensive evaluation was for yaw/pitch/roll image stabilization performance. Evaluation is performed at the center where, although the image stabilization performance of roll cannot be evaluated, many main subjects are placed, and evaluation is performed at the corner to evaluate the image stabilization performance of the three components yaw, pitch, and roll combined.

Information such as the camera name, lens name, and focal length is added in * as necessary.

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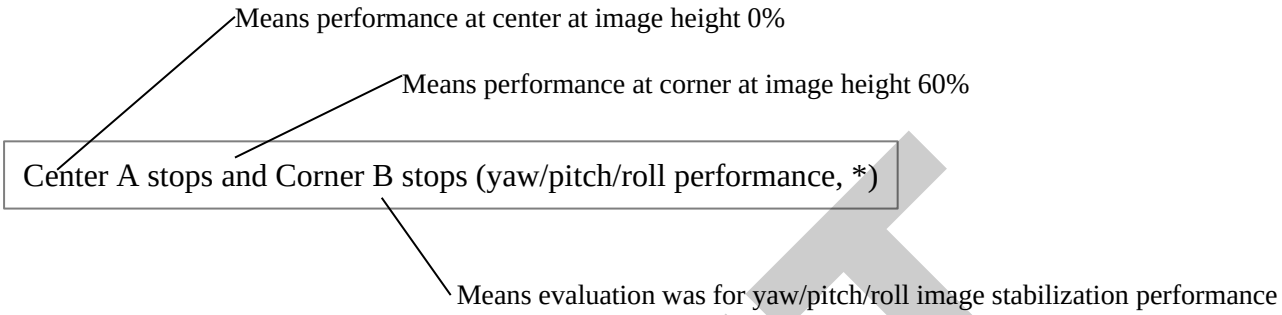


Figure 6-25a Explanation of Wording in Performance Description (1)

The main section of the performance description example of a single imaging lens is Center A stops (yaw/pitch performance, *).

Center means performance at center at image height 0%.

The “yaw/pitch performance” written after “Center A stops” means that the evaluation was for yaw/pitch image stabilization performance and conducted only at the center without a roll handheld blur component. General image stabilization mechanisms built into imaging lenses can compensate for yaw/pitch and shift handheld blur, but cannot compensate for roll handheld blur. Therefore, only the center was evaluated and not the corner positions.

Information such as the camera name, lens name, and focal length is added in * as necessary.

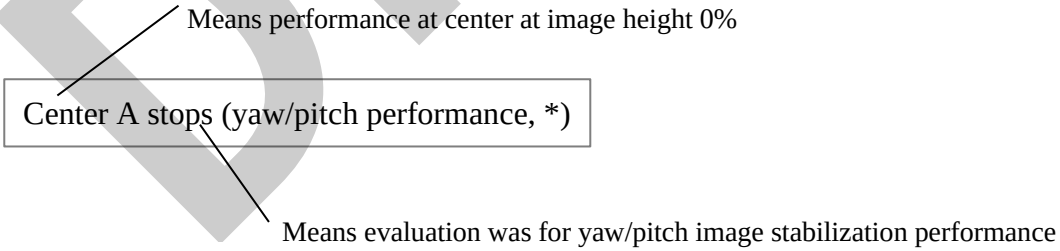


Figure 6-25b Explanation of Wording in Performance Description (2)

Variations of Performance Description Examples in Brochures

Description examples are provided in 5. Description Method. The following are variations based on

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the description examples. These variations, however, are merely examples and not limitations.

Variation of 5-3-1 Camera Body with Image Stabilization in Roll Direction of 5-3 Camera Body with Image Stabilization Mechanism:

Example of specifying all contents in specification table:

Specification Table of Camera B	
XX	XX
Image stabilization	yaw/pitch/roll/shift image stabilization
Image stabilization performance	Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 24-70mm lens, f = 70mm, IS auto of image stabilization mode)
YY	YY

Example specifying contents separately in specification table and description:

Specification Table of Camera B	
XX	XX
Image stabilization performance	Center 5 stops and Corner 4 stops (yaw/pitch/roll performance, 24-70mm lens, f = 70mm, IS auto of image stabilization mode)
YY	YY

In description:
. . . This camera B is equipped with a 5-axis image stabilization mechanism . . .

Example of specifying all contents in description:

In description:
. . . This camera B is equipped with a 5-axis image stabilization mechanism.
. . . The image stabilization performance: Center 5 stops and Corner 4 stops with a 24-70mm lens (yaw/pitch/roll performance, f = 70mm, IS auto of image stabilization mode) . . .

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Variation of 5-4-1 Interchangeable Lens with Image Stabilization in Yaw/Pitch Directions of 5-4 Interchangeable Lens with Image Stabilization Mechanism

Example of specifying all contents in specification table:

Specification Table of Lens A	
XX	XX
Image stabilization	Yaw/pitch image stabilization
Image stabilization performance	Center 5 stops (yaw/pitch performance, f = 70mm, IS auto of image stabilization mode with camera B)
YY	YY

Example specifying contents separately in specification table and description:

Specification Table of Lens A	
XX	XX
Image stabilization performance	Center 5 stops (yaw/pitch performance, f = 70mm, IS auto of image stabilization mode with camera B)
YY	YY

In description:
. . . This lens A is equipped with a 2-axis image stabilization mechanism (yaw/pitch image stabilization) . . .

Example of specifying all contents in description:

In description:
. . . This lens A is equipped with a 2-axis image stabilization mechanism.
. . . The image stabilization performance: Center 5 stops with camera B (yaw/pitch performance, f = 70mm, IS auto of image stabilization mode) . . .

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Explanation of Performance Description Examples in Brochure of Interchangeable Lens

In a brochure of an individual interchangeable lens, only the number of stops of image stabilization performance at the center is described in principle in accordance with the provisions of 5-4 Interchangeable Lens with Image Stabilization Mechanism of 5 Description Method. However, the number of stops of image stabilization performance at the center and at image height 60% may be indicated in accordance with the provisions of 5-5 Pairing of Camera Body with Image Stabilization Mechanism and Interchangeable Lens with Image Stabilization Mechanism.

Two examples of such descriptions are given below.

Example specifying contents separately in specification table and description:

Specification Table of Lens A	
XX	XX
Image stabilization	Yaw/pitch image stabilization
Image stabilization performance	Center 5 stops (yaw/pitch performance, f = 70mm, with camera B)
YY	YY

In description:
. . .
When this lens A is paired with camera C (equipped with yaw/pitch/roll image stabilization mechanism), the performance is Center 5.5 stops and Corner 4.5 stops (yaw/pitch/roll performance, f = 70 mm),
When this lens A is paired with camera D (equipped with 5-axis image stabilization mechanism), the performance is Center 6 stops and Corner 5 stops (yaw/pitch/roll performance, f = 70 mm),
. . .

Example of specifying all contents in specification table:

Specification Table of Lens A	
XX	XX
Image stabilization	Yaw/pitch image stabilization

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Image stabilization performance	Center 5 stops (yaw/pitch performance, f = 70mm, with camera B), Center 5.5 stops and Corner 4.5 stops (yaw/pitch/roll performance, f = 70mm, with camera C (equipped with yaw/pitch/roll image stabilization mechanism)), Center 6 stops and Corner 5 stops (yaw/pitch/roll performance, f = 70mm, with camera D (equipped with 5-axis image stabilization mechanism))
YY	YY

Example of Use of Performance Description

If the number of stops of image stabilization performance are listed as Center A stops and Corner B stops, the photographer can take advantage of the two stop values.

For example, if photographing a person at the center of the image and not concerned about handheld blur in the peripheral background or an out-of-focus periphery, the photographer can set the shutter speed with reference to the number of stops of image stabilization performance at the center (A) and take the photograph with minimal handheld blur.

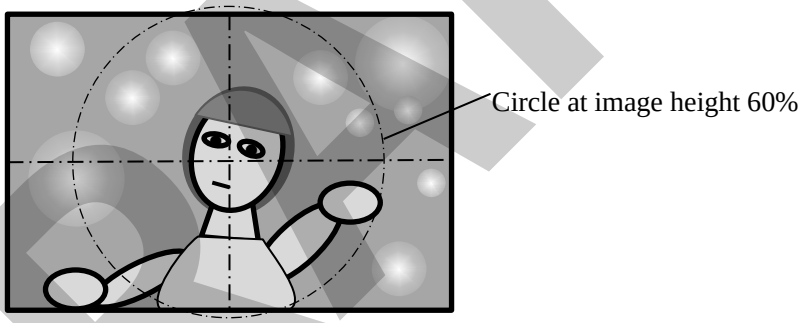
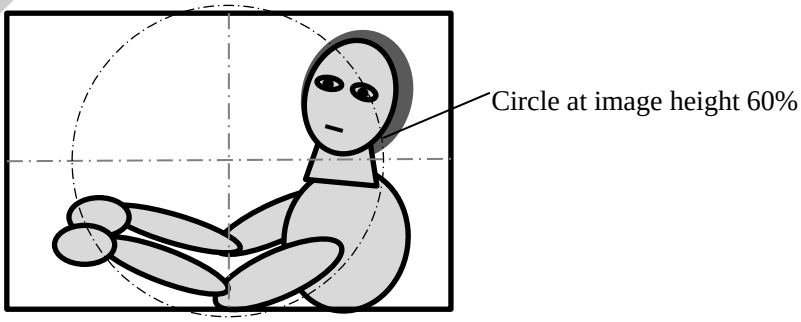


Figure 6-25c Photographing a Person (with Eyes Substantially at Center, and Periphery Out-of-Focus)

If photographing a person at the corner of the screen, the photographer can set the shutter speed with reference to the number of stops of image stabilization performance at the corner (B) and take the photograph with minimal handheld blur.



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Figure 6-25d Photographing a Person (with Eyes at Image Height 60%)

In landscape photography in which the entire screen is in focus, it is often preferred to take the photograph with minimal handheld blur both at the center of the image and the corner of the image. In this case, the photographer can set the shutter speed with reference to the number of stops of image stabilization performance at the center (A) and at the corner (B) and take the photograph across a wide range of the screen with minimal handheld blur.

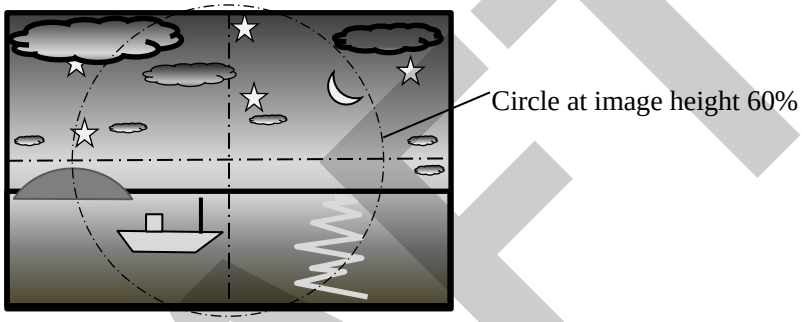


Figure 6-25e Photographing a Landscape (with Entire Screen in Focus at Infinity)

In this way, the photographer can set an appropriate shutter speed that does not cause noticeable handheld blur of the main subject by using two different numbers of stops of image stabilization performance on the basis of the position (image height) of the main subject.

6-26 Action when Number of Stops at Image Height 60% Is Not Measurable

With a super-telephoto lens of about 800mm (35mm film equivalent), for example, when vibration is started after the center of the image is aligned with the center of the CIPA handheld blur measurement chart, an image may be captured with the center of the image and the center of the chart significantly misaligned. As a result, the position at image height 60% on the chart may be off-screen or close to the edge of the screen, making measurement not possible.

This occurs when the image stabilization operation of the lens or camera is not aligned with the low-frequency, large-amplitude operation of the vibration table. The diagonal measurement positions at image height 60% on the chart tend to be close to the screen edge at the maximum amplitude of pitch handheld blur. The maximum amplitude of pitch handheld blur of the vibration waveform TWB-H is approximately ±0.4°. With an image aspect ratio of 3:2, the angles (margin angles) from the

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diagonal positions at image height 60% to the top and bottom edges of the image are calculated to be 1.4 degrees at a focal length of 200mm, 0.7 degrees at 400mm, 0.5 degrees at 600mm, and 0.4 degrees at 700mm. Depending on the image stabilization characteristics and exposure timing, the focal length of a lens with a margin angle of less than the maximum amplitude of approximately ±0.4 degrees is 700 mm (margin angle: 0.4 degrees) or greater, possibly causing the chart diagonal measurement positions at image height 60% to be off-screen or close to the screen edge, making measurement not possible.

In this state, the performance descriptions specified in 5-2 Performance Description of Lens Integrated Camera with Image Stabilization Mechanism, 5-3 Performance Description of Camera Body with Image Stabilization Mechanism, and 5-5 Performance Description of Pairing of Camera Body with Image Stabilization Mechanism and Interchangeable Lens with Image Stabilization Mechanism cannot be listed.

On the other hand, with a focal length of 200mm (margin angle: 1.4 degrees) or less, there is enough margin for a maximum amplitude of approximately ±0.4 degrees, never resulting in chart diagonal measurement positions at image height 60% being off-screen or close to the screen edge and the consequential impossibility of measurement.

Examples of Action when Number of Stops at Image Height 60% Is Not Measurable

Example 1:

If measurement is not possible at image height 60%, measurement can often be carried out by setting the focal length of the lens to be measured to a wider angle.

Example 2:

When 4-4-4 Action when Total Image Degradation Amount at Image Height 60% Is Not Measurable of 4. Measurement Method is applicable, photographs not measurable at image height 60% may be excluded. In this case, additional photographs may be taken until at least 200 measurable photographs are obtained for selection criteria I and II, or additional photographs may be taken until at least 100 measurable photographs each are obtained for selection criteria III using two kinds of waveforms, and the total image degradation amount may be calculated using those photographs.

However, if the number of additional shots is large, the measurement at image height 60% may be canceled.

If the measurement is canceled, only the number of stops of image stabilization performance at the center of the image shall be listed, and the description shall clearly indicate that the number of stops at image height 60% was not measured, in accordance with 5. Description Method.

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6-27 Difference in Number of Stops of Image Stabilization Performance between Center and Corner

The number of stops of image stabilization performance tends to be lower at the corner than at the center. The amount of reduction in the number of stops at the corner relative to the center tends to be smaller for telephoto lenses than for wide-angle lenses. The reason is as follows.

For ease of explanation, description will be made using a wide-angle lens having focal length of 24mm and a telephoto lens having focal length of 200mm as examples.

In principle, there is no roll handheld blur effect at the center. Therefore, the technical difficulty of reducing the 200μm combined yaw and pitch handheld blur amount caused by vibration waveforms to the handheld blur determination level of 20μm, that is, to 1/10, is about the same for both the 24mm wide angle and the 200mm telephoto.

On the other hand, at the corners (image height 60%), there is a roll handheld blur effect, and the shutter speed corresponding to a 200μm combined yaw, pitch, and roll handheld blur amount caused by vibration waveforms is approximately 1.5 seconds at 24mm and approximately 1/19 seconds at 200mm. At this time, the ratio of the roll handheld blur amount to the combined yaw, pitch, and roll handheld blur amount and the roll handheld blur amount are 23% and approximately 57μm at 24mm and 3% and approximately 7μm at 200mm.

The yaw and pitch handheld blur amounts are values close to 200μm, and thus the technical difficulty of reducing these amounts to 1/10 is the same as that at the center described above.

However, compared to the yaw and pitch handheld blur amounts, the roll handheld blur amounts are as small as 57μm and 7μm, and the technical difficulty of further reducing these small amounts to 1/10 is high.

Therefore, even if it is technically possible to reduce the yaw and pitch handheld blur amounts to 1/10, it may not be possible to reduce the roll handheld blur amount to 1/10. As a result, the number of stops of image stabilization performance tends to be lower at the corners than at the center.

The roll handheld blur amounts are 57μm at 24mm and already a minute 7μm at 200mm. Therefore, even if the handheld blur amount cannot be reduced to 1/10, the amount of residual handheld blur is very small. Thus, with the roll handheld blur amount at 200mm being small to begin with, the residual handheld blur is very small even if the blur reduction effect is small. Accordingly, at 200mm, compared to 24mm, the reduction in the number of stops of image stabilization performance at the corners tends to be small compared to that at the center.

In other words, the yaw and pitch handheld blur amounts increase in proportion to focal length, but the roll handheld blur amount does not change. As the focal length becomes increasingly telephoto, the shutter speed to be measured increases and the yaw and pitch handheld blur amounts and roll

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handheld blur amount are reduced by the same ratio. Therefore, the roll handheld blur effect is relatively small, and the difference between the numbers of stops at the center and at the corner tends to decrease in the telephoto range.

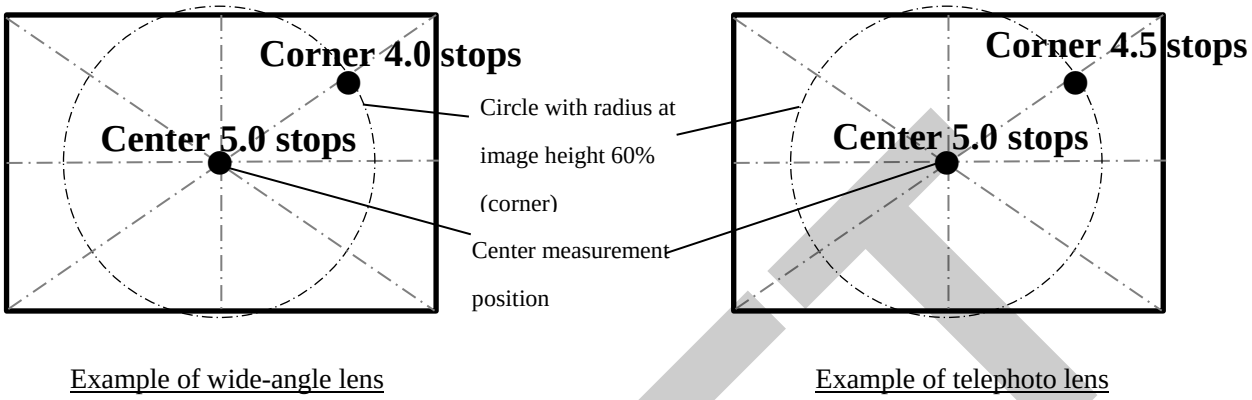


Figure 6-27 Example of Difference in Number of Stops of Image Stabilization Performance

6-28 Vibration Waveforms

The 2024 standard DC-X011-2024 revision adds a roll motion blur component to the vibration waveforms, resulting in the vibration waveforms having the three rotating components of yaw, pitch, and roll. Camera shake caused by the human hand includes components in six axes total (yaw, pitch, roll, X, Y, and Z), but the three X, Y, and Z translational components are not addressed herein. Camera shake has components in six axes total (yaw, pitch, roll, X, Y, and Z).

Ideally, the vibration waveforms should be similar to those of camera shake caused by the human hand. Using advanced measurement technology, CIPA measured the vibration of a handheld camera, analyzed the roll vibration component, and successfully standardized the roll vibration waveform for the vibration table in this revision. This standard makes it possible to induce vibration and take measurements in a state closer to the camera shake caused by the human hand than the 2015 standard DC-001-2015 by using three-axis vibration including yaw and pitch, which are particularly dominant vibration components of the captured image, and roll, a component having the next largest impact.

Reason for Measurement Based on Vibration of All Three Axes

The first reason for measurement based on all three axes is to simulate camera shake in a state closer to that caused by the human hand. Therefore, regardless of the image height of measurement, vibration on the three axes of yaw, pitch, and roll shall be required.

The second reason is to further include evaluation of roll image stabilization. This evaluation

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requires measurement at a certain image height, such as image height 60%, with vibration on the three axes of yaw, pitch, and roll.

On the other hand, at the center of the image, roll handheld blur does not occur in principle, resulting in the expectation that the number of stops of image stabilization performance would be the same even if the measurement thereof at the center of the image was simplified to measurement based on vibration on the two axes of yaw and pitch instead of the three axes of yaw, pitch, and roll. However, actual measurement, at times, told a different story.

One possible factor is the overall difficulty of image stabilization. That is, when only yaw and pitch vibrations are used and not roll vibration, it is only necessary for the sensor to separately detect yaw and pitch motion blur and for the image stabilization mechanism to stabilize the two yaw and pitch components. However, in the case of vibration with the three components of yaw, pitch, and roll, it is necessary for the sensor to separately detect the yaw, pitch, and roll components of the camera shake and for the image stabilization mechanism to stabilize the three components of yaw, pitch, and roll.

In this standard, therefore, regardless of whether image stabilization performance at image height 60% is measured or not, the three vibration waveforms of yaw, pitch, and roll shall be simultaneously applied during all measurements.

Explanation of Shift Handheld Blur

Current cameras are sometimes equipped with a mechanism configured to reduce handheld blur translating in two axial directions within a plane orthogonal to the optical axis. This handheld blur is referred to as "shift handheld blur."

The impact of shift handheld blur appears at imaging distances closer than about 20 times the 35mm film equivalent focal length, and is even greater at magnifications close to 1x (equal magnification) with lenses called macro lenses or micro lenses.

Why Shift Image Stabilization Was Not Standardized in this Revision

In order to properly evaluate shift image stabilization, the shift vibration waveform and imaging magnification shall be appropriately set. With the distance being so close, the measurement accuracy is greatly affected by the positioning accuracy of the camera on the vibration table, the positioning accuracy of the center of rotation of the vibration table, and the positioning accuracy and squareness of the chart. In addition, changes in focus during vibration also impact measurement accuracy.

However, at this time, without an evaluation method properly satisfying these conditions, this standard does not incorporate a method for measuring shift image stabilization performance.

6-29 Difference in Number of Stops of Image Stabilization Performance Between 2015 Standard

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(CIPA DC-011–2015) and 2024 Standard (CIPA DC-X011-2024)

In the 2015 standard (CIPA DC-011-2015), yaw and pitch vibration waveforms were used to measure the number of stops of image stabilization performance at the center of the image at a handheld blur determination level of 63μm.

In the 2024 standard (CIPA DC-X011-2024), the roll vibration waveform is newly added to the yaw and pitch vibration waveforms of the 2015 standard to measure the number of stops of image stabilization performance at the center of the image and at image height 60% at a handheld blur determination level of 20μm.

With the handheld blur determination level being tightened from 63μm to 20μm, the amount of handheld blur in an image shall be reduced on the camera to 20μm or less, possibly changing the number of stops of image stabilization performance measured with the camera.

Further, with a roll vibration waveform being added to the yaw and pitch vibration waveforms, the three vibration components shall be accurately separated into the yaw, pitch, and roll vibration components and detected on the camera. Thus, while the 2015 standard only required the separate detection of two components, the 2024 standard requires separate detection of three components, which increases technical difficulty and, as a result, is likely to reduce the number of stops of image stabilization performance.

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7. Participating members

The bulk of the deliberations over the formulation of the standards described in this document was performed by the Anti-blur Sub-Working Group.

The members of the Sub-Working Group are listed below;

Standardization Committee

Technical Working Group

Anti-blur Sub-Working Group

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Appendix (Informative)

Explanation of Terminology Relating to Handheld Blur for Digital Cameras

A-1 Introduction

CIPA has issued the “CIPA DCG-002-2012 Specification Guideline for Digital Cameras (Revised Version)” as a guideline for describing the specifications of digital cameras in brochures and the like. The document includes definitions of terminology relating to handheld blur, and how it should be used. Such terminology includes “image stabilization” and “blur reduction,” and also “blur suppression,” which is a generic term encompassing the former two expressions.

However, at present, it is possible to find examples in brochures and on the websites of suppliers where such expressions relating to handheld blur have been defined differently or without clear distinctions between them. Aside from causing confusion for users, such misuse of terminology prevents fair competition between rival enterprises.

For the above reason, this Appendix seeks to systematically reorganize the terminology relating to handheld blur used for digital cameras and provides a definition-based explanation.

The same information in this appendix also forms the content of the appendix of “CIPA DCG-007-2015 Description method for Image Stabilization Performance of Digital Cameras (Electronic/Hybrid Systems).”

A-2 Basic concept of this Appendix

“CIPA DCG-002-2012 Specification Guideline for Digital Cameras (Revised Version)” contains “Commentary (6)-3. Further clarifications of ‘image stabilizer’ and ‘blur reduction’,” in which examples are provided of how “(a) image stabilizer” and “(b) blur reduction {mode}” as described in “Item 17: Blur Suppression” are respectively used. This appendix reproduces Commentary (6)-3 but provides more, detailed explanation.

A-3 Explanation

In line with the establishment of “CIPA DCG-002-2012 Specification Guideline for Digital Cameras (Revised Version),” “CIPA DCG-007-2015 Description Method for Image Stabilization Performance of Digital Cameras (Electronic/Hybrid Systems),” and this present standard, further clarification is

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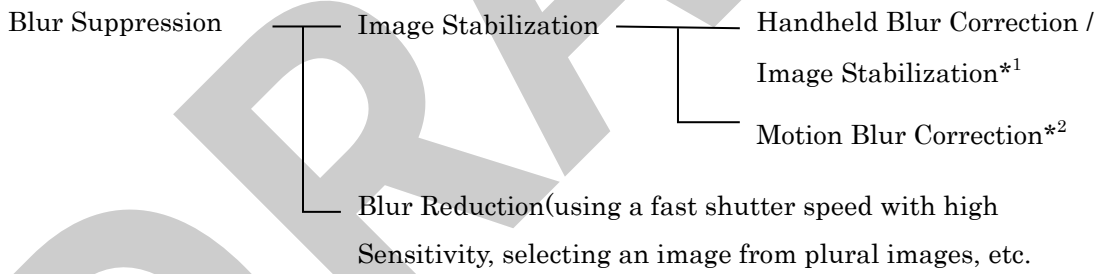
given of the definitions of “image stabilization” and “blur reduction.” Also indicated are “terminology relating to blurring,” examples of “output by blur detecting devices,” definition of the “image stabilizer” function and examples, the definition of “blur reduction” and examples, cases where the image stabilizer function has a different method and subject, and when both the image stabilizer function and blur reduction are present.

A-3-1 Terminology relating to blurring

“Blur” is interpreted as displacement from a given position, and when used as camera terminology, blur can be broadly classified into “handheld blur” and “motion blur”.

“Handheld blur” refers to blurring of the subject in shot images as a result of camera movement due to instability of the hand(s) holding the camera. “Motion blur” refers to blurring of the subject in shot images as a result of movement of part or all of the subject during shooting irrespective of whether or not there is handheld blur.

Note that although “vibration isolation” is known as a method of preventing shake, “vibration isolation” in used in regard to equipment that prevents vibration from being applied to the camera itself, and is therefore terminology from a different field to image stabilization.



*1 Optical systems(VAP, lens shift system, sensor shift system, etc.
Electronic systems(image synthesis image restoration, etc.)
Hybrid systems(concurrently using an optical system and an electronic system)

*2Electronic systems(image synthesis, tec.)

The tree diagram given above shows terminology relating to blur. The following content about blur suppression is given in the “CIPA DCG-002-2012 Specification Guideline for Digital Cameras (Revised Version),” but here contains more, detailed explanation.

A-3-1-1 Blur suppression

The expressions “image stabilizer” and “blur reduction” are generic terms and refer to

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the suppression of “handheld blur” and/or “motion blur” by some kind of control means provided in a camera. “Blur suppression” is broadly classified into “image stabilizer” and “blur reduction”.

a) Image stabilizer

This refers to a function that uses the output by a blur detection device to correct blurring in output images caused by movement of the camera itself (due to handheld blur or the like) and/or movement of the subject. However, even if the output by a blur detection device is used, systems where shooting is carried out using a fast shutter speed as a result of optimization by an exposure control program and systems where no significant improvement is observed in output images in which handheld blur has been corrected are regarded as b) “blur reduction.”

The category “image stabilizer” is divided into “handheld blur correction” and “motion blur correction.” Optical, electronic and hybrid systems exist for “handheld blur correction.”

b) Blur reduction

This refers to a function or mode that reduces blurring in output images caused by movement of the subject and/or handheld blur and its main means is using a fast shutter speed as a result of optimization by an exposure control program. Although blur reduction is one method of an exposure control mode and a sensitivity control mode, there are no problems with this function being separately referred to as a blur reduction function or mode.

A-3-2 Examples of “output by blur detection device”

a) Measurement using a gyro-sensor or the like.
b) Measurement by comparing images from continuous shooting (motion vectors between continuously shot images or the like)
c) Measurement through analysis of blur (image degradation) in single images (calculation of the blur (image degradation function))

A-3-3 Explanation of “image stabilizer” and examples

A-3-3-1 Explanation

The image stabilizer function is interpreted here to mean a function that uses the output by a blur

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detection device to correct blurring in output images caused by movement of a camera itself (due to handheld blur or the like) and/or movement of the subject. Accordingly, if there is significant deterioration in image quality as a side effect of correction when no significant correcting effect has been achieved, or when the probability of a correcting effect being obtained is extremely low, it is not possible to regard the technology being used as “image stabilizer” technology.

A-3-3-2 Optical, electronic, and hybrid systems

Out of the examples of the image stabilizer function given below in A-3-3-3, a) and b) are called optical systems, while c) to e) are called electronic systems. Systems that use a combination of optical and electronic technology are called “hybrid systems” and usually have characteristics of both types of technology. Examples of hybrid systems are listed below.

・ Systems where optical and electronic technology operates simultaneously.

・ Systems that appropriately switch between optical and electronic technology according to the shooting conditions and the extent of the handheld blur.

・ Systems where correction according to an optical system and correction according to an electronic system are carried out on different axes, with both systems being present.

A-3-3-3 Examples of systems that are considered to be suitable for classification as “image stabilizer” systems. (NB. These are just examples, and the image stabilizer function is not limited to such.)

a) A method where subject displacement in the image caused by handheld blur is corrected by moving part or all of the lens in parallel relative to the image sensor using the output by a blur detection device.

b) A method where subject displacement in the image caused by handheld blur is corrected by rotating or moving in a perpendicular direction relative to the optical axis using the output by a blur detection device.

c) A method where multiple images with reduced blur are taken at a fast shutter speed and such images are combined (aligned etc.) using the output by a blur detection device to obtain an output image with reduced blur and virtually the same image quality (S/N) as an image shot at the usual shutter speed.

d) A method where an image with reduced blur shot at a fast shutter speed and an image shot at the usual shutter speed are combined (aligned etc.) using the output by a blur detection device to obtain an output image with reduced blur and virtually the same image quality (S/N) as an image shot at just the usual shutter speed.

e) A method that is based on processing using a point spread function, an inverse filter or the like

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and uses information output by a blur detection device to restore an image with little blur from a shot image.

A-3-4 Explanation of “blur reduction” and examples

A-3-4-1 Explanation

Blur reduction is interpreted here to mean a function or mode of reducing blurring in output images that is caused by movement of the subject and/or handheld blur. It is mainly achieved using a fast shutter speed as a result of optimization by an exposure control program. In this way, since “image stabilization” and “blur reduction” have very different technical aspects, such expressions should not be confused.

A-3-4-2 Characteristics of “blur reduction”

Example characteristics of blur reduction are given below.

Merits: Since blur reduction can be achieved simply by optimization of an exposure control program, implementation is simple and such technology is also effective against motion blur.

Demerits: Since high ISO sensitivity is required in many cases, there is increased noise in the shot images.

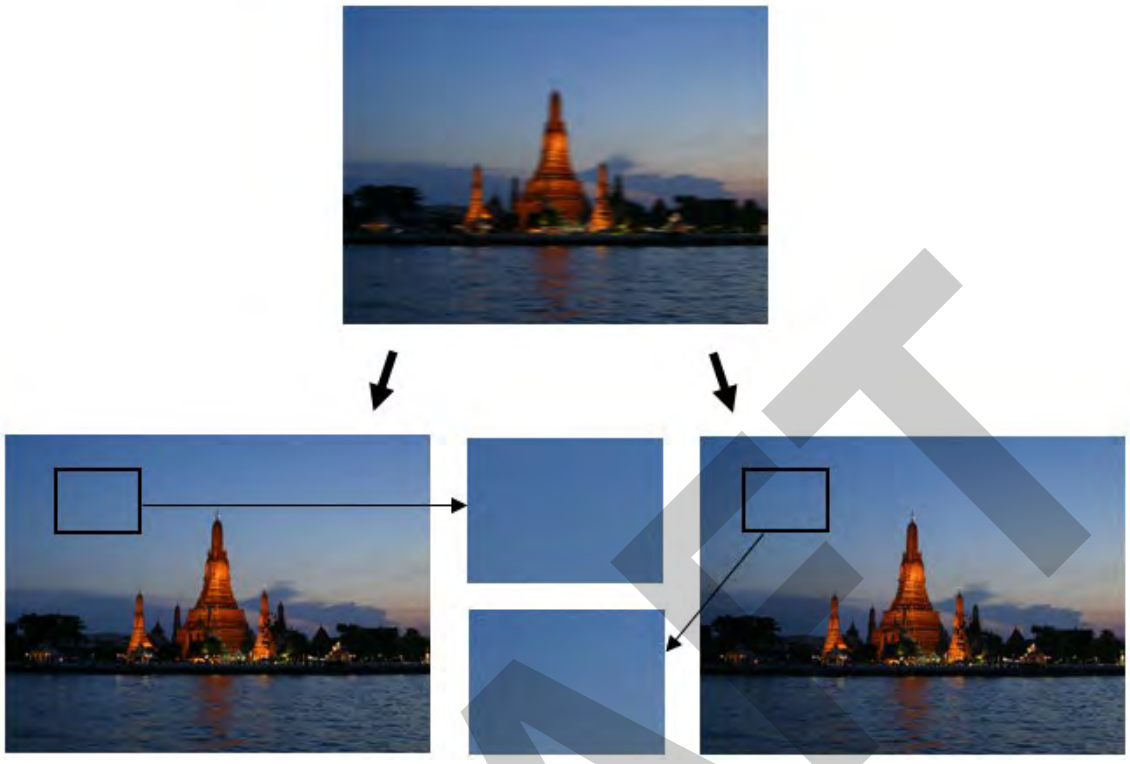
A-3-4-3 Difference in image quality between “image stabilizer” and “blur reduction”

Fig. 1 shows an example comparison between the blur suppression effects of the image stabilizer function and blur reduction. Although setting the functions of image stabilizer and blur reduction to ON results in a similar reduction in handheld blur, an increase in noise is observed in the image with blur reduction.

In this way, since blur reduction is a method that suppresses handheld blur while mainly sacrificing image quality (and S/N in particular), it is appropriate to regard blur reduction as having no image stabilization effect. Accordingly, when measuring and describing image stabilization effects, it is necessary to pay particular attention to excluding effects achieved by blur reduction.

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Fig. 1. Difference in image quality between image stabilization and blur reduction



A-3-4-4 Examples of systems that are considered to be suitable for classification as "blur reduction" (NB. These are just examples, and blur reduction is not limited to such.)

a) A method that sets sensitivity higher than in the usual shooting mode, shoots an image with a higher-than-usual shutter speed, and carries out image processing such as edge enhancement to compensate for loss of resolution in the shot image.

b) A method that sets sensitivity higher than in the usual shooting mode, shoots an image with a higher-than-usual shutter speed, and carries out image processing such as noise reduction to compensate for S/N deterioration in the shot image.

c) A method that sets sensitivity higher than in the usual shooting mode and shoots an image with a higher-than-usual shutter speed when motion detection for the subject has found that all or part of the subject is moving at a given speed or faster.

d) A method that sets sensitivity higher than for the usual shooting mode, takes multiple shots with a higher-than-usual shutter speed, and records only images that the camera has determined to have the least blurring.

e) A method that shoots multiple images while gradually changing the sensitivity and shutter speed, and records only images that the camera has determined to have the best balance between the amount of blur and S/N.

f) A method that shoots at both the usual shutter speed and a higher-than-usual shutter speed, and

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records the image shot at the usual shutter speed if there is little blurring, or records the image shot at the higher-than-usual shutter speed if there is significant blurring with the usual shutter speed.

A-3-5 Objective axis-dependent and still- or video-dependent image stabilization

In a camera equipped with a hybrid image stabilizer function, if such camera has several axes where correction is carried out optically and other axes where correction is carried out electronically, this should be clearly indicated to users (see a) below). If the image stabilizer function is also enabled during the shooting of video images, when the correcting system or objective axes for correction are different to those used in shooting still images, care should be taken to clearly indicate such to users (see b) below).

a) Yaw/pitch is optically corrected, while up-down, left-right, and rotation are electronically corrected.
b) Correction is carried out on two axes (yaw and pitch) when shooting still images and correction is further carried out on the up-down, left-right, and rotational axes (i.e., a total of five axes) when shooting video images.

A-3-6 Combined use of image stabilizer function and blur reduction

As described earlier, since the effect of the image stabilizer function and the effect of blur reduction cannot be directly compared, when measuring the effect of the electronic image stabilizer function, it is necessary to completely eliminate the effect of blur reduction. However, since there are many cameras that exhibit the properties described in a) and b) below, sufficient attention needs to be paid when measuring and describing the effect of the image stabilizer function.

a) A configuration that functions only as an image stabilizer when there is little blurring, but increasingly functions as a handheld blur reducer according to blurring level. It functions as only a handheld blur reducer at or above a given level of blurring.
b) A configuration that functions as an image stabilizer and handheld blur reducer when there is little blurring, but functions as only a handheld blur reducer at or above a given level of blurring.

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